

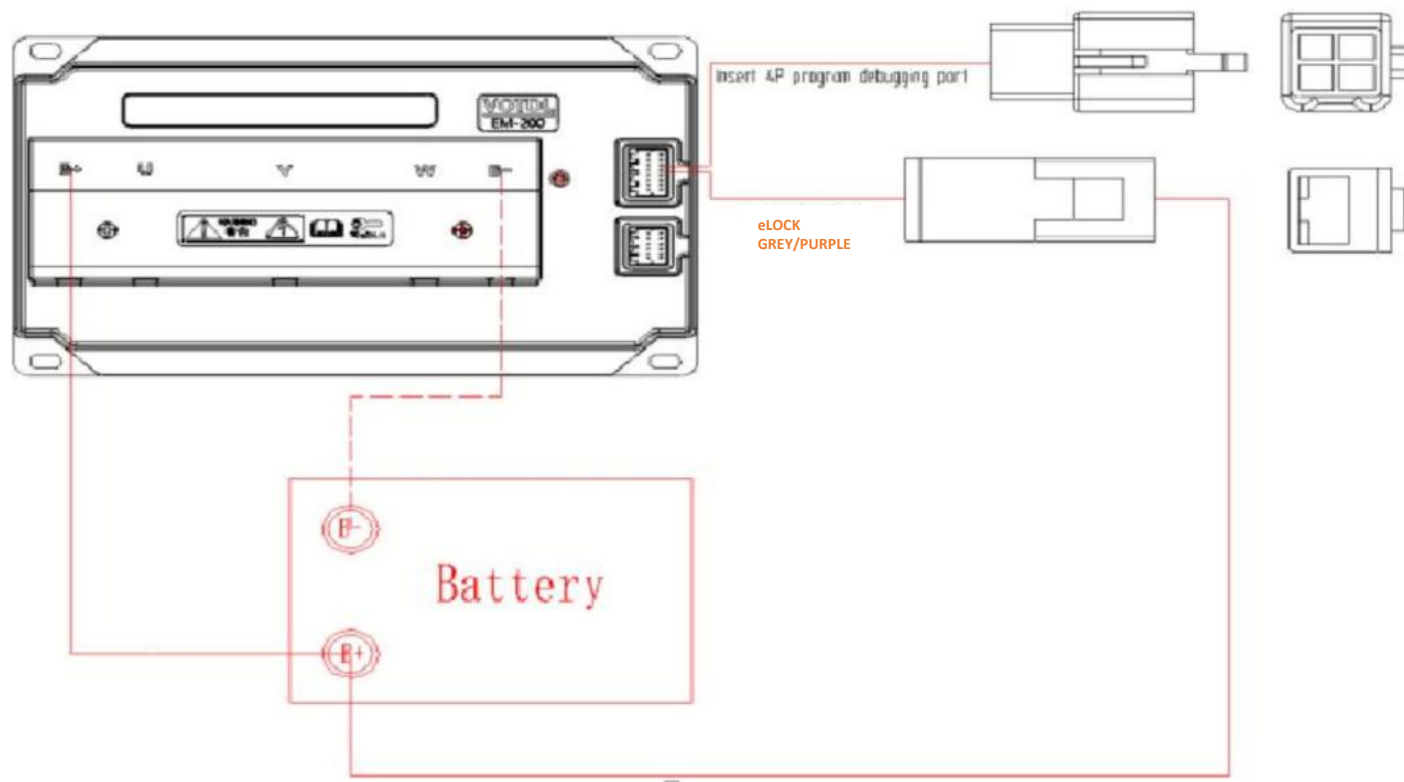
PROGRAMMING MANUAL OF THE VOTOL CONTROLLER

1

INITIAL CONNECTIONS

To program the controller, you must have the following power supplies active

The controller must be powered up by the power supplier line(Gray/Purple) with Elock+ (above 35V) for debugging .



VOTOL-EM-V3 Software Debugging

PAGE1 | PAGE2 | PAGE3 | Display | Port Settings

SW: 0 HW: 0 buadrate: 115200 ☒ CAN enable ÖĐİÄ

Basic Settings

Model:

Battery Voltage(V):

Overvoltage(V):

Undervoltage(V):

Soft undervoltage(V):

Undervoltage variation:

Busbar current(A):

Phase current:

throttle voltage set(max. 5.5v)

low protect:

start voltage:

the end of the high:

start setting

start torque:

combinative rate of rise:

rate of decline:

PORT: COM3 success_count: 0 error_count: 0

status:

Annotations:

- Activate this box for programming (which is done via CanBus) Tick for CAN, Untick for LIN
green box with 4.0 version do not need ✓
- Commucation speed setting
通讯速度设置
CAN base choose 11520
- Choose the USB connection port and click on the “OPEN” button. It will go to “CLOSE” state and then click on “CONNECT”. The parameters will pop out once it is connect successfully. It may pop out abnormal connection or unable to connect if the USB cable is not connected well or the controller is not powered on.
- Press this button after implementing changes to the parameters.
- Import a saved parameter configuration(ini file).
- Save the created parameter configuration(bin file).

Basic Settings

Model: EM-100

Battery Voltage(V): 76V

Indicate the controller to be programmed and define the nominal voltage of the battery pack.
model from EM30 to EM200, all can be chosen

Overvoltage(V): 91,20

Undervoltage(V): 67

Indicate the maximum and minimum values allowed (according to the battery parameters)

Default +2V for over voltage select, while default +1V for under voltage select

VOTOL-EM-V3 Software Debugging

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SW: 0 HW: 0 buadrate: 115200 CAN enable

Basic Settings

Model: Battery Voltage(V):

Overvoltage(V): 0

Undervoltage(V): 0

Soft undervoltage(V): 0

Undervoltage variation: 0

Busbar current(A): 0

Phase current: 0

throttle voltage set(max. 5.5v)

low protect: 0

start voltage: 0

the end of the high: 0

start setting

start torque: 0

combinative: 0

rate of rise: 0

rate of decline: 0

Attention:

- 1, The controller will not output if it detect the voltage is higher than the overvoltage value or lower than the undervoltage value. It will report the fault code accordingly.
- 2, The value of undervoltage and overvoltage must be set according to the battery spec to avoid the hazard the battery cell and BMS components.

Soft undervoltage:

The controller determines the current voltage value, which is 3V higher than the battery undervoltage value. If the current voltage value is lower than the soft undervoltage value, the controller output is immediately switched off to protect the battery.

VOTOL-EM-V3 Software Debugging

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SW: 0 HW: 0 baudrate: 115200 CAN enable

Basic Settings

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Undervoltage(V):

Soft undervoltage(V):

Undervoltage variation:

Busbar current(A):

Phase current:

throttle voltage set(max. 5.5v)

low protect

start voltage:

the end of the high

start setting

start torque:

combinative

rate of rise:

rate of decline:

Undervoltage variation:

Para evitar medidas incorrectas por parte del controlador, se incrementa el valor de la tensión "undervoltage" para no provocar un daño en el pack de baterías. Normalmente se suele marcar 1V, pero es posible aumentar dicho valor si queremos tener más seguridad.

To avoid incorrect measurements by the controller, the value of the "undervoltage" voltage is increased so as not to cause damage to the battery pack. **Normally set with 3V**, but it is possible to increase this value if we want to have more security.

Busbar current(A):	30
Phase current:	9600

Busbar Current



VOTOL-EM-V3 Software Debugging

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Basic Settings

Model:	
Battery Voltage(V):	
Overvoltage(V):	0
Undervoltage(V):	0
Soft undervoltage(V):	0
Undervoltage variation:	0
Busbar current(A):	0
Phase current:	0

throttle voltage set(max. 5.5v)

low protect	0
start voltage:	0
the end of the high	0

start setting

start torque:	0
combinative	0
rate of rise:	0
rate of decline:	0

Phase Current Limited Value

Phase current = 9600

Especificar este valor y no modificar.

Specify this value and do not modify.

VOLTAGE MODEL	EM-30S	EM-50	EM-50S	EM-100	EM-100S	EM-150	EM-150S	EM-20	EM-30
48-60V	33A	45A	50A	85A		120A	150A	200A	400A
72V	33A	45A	50A	85A	100A	120A	180A	200A	400A
84V	30A	45A		80A		120A			
96V		40A		70A		100A		180A	350A

Busbar current(A):

Phase current:

Busbar Current

Corriente máxima permitida de alimentación al controlador.

El límite del controlador es 100A (72V)

Potencia máxima = 72V x 100A = 7200W = 7,20kW

Maximum allowable power supply current to the controller. Controller limit is 100A (72V)Maximum power = 72V x 100A = 7200W = 7.20kW

VOTOL-EM-V3 Software Debugging

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Basic Settings

Model:

Battery Voltage(V):

Overvoltage(V):

Undervoltage(V):

Soft undervoltage(V):

Undervoltage variation:

Busbar current(A):

Phase current:

throttle voltage set(max. 5.5v)

low protect

start voltage:

the end of the

high

start setting

start torque:

combinative

rate of rise:

rate of decline:

Phase Current Limited Value

Phase current = 9600

Especificar este valor y no modificar.

Specify this value and do not modify.

VOTOL-EM-V3 Software Debugging

PAGE1 PAGE2 PAGE3 Display Port Settings

SW: 0 HW: 0 buadrate: 115200 ☒ CAN enable

Basic Settings

Model:

Battery Voltage(V):

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Undervoltage(V):

Soft undervoltage(V):

Undervoltage variation:

Busbar current(A):

Phase current:

throttle voltage set(max. 5.5v)

low protect

start voltage:

the end of the high

start setting

start torque:

combinative

rate of rise:

rate of decline:

throttle voltage set(max. 5.5v)

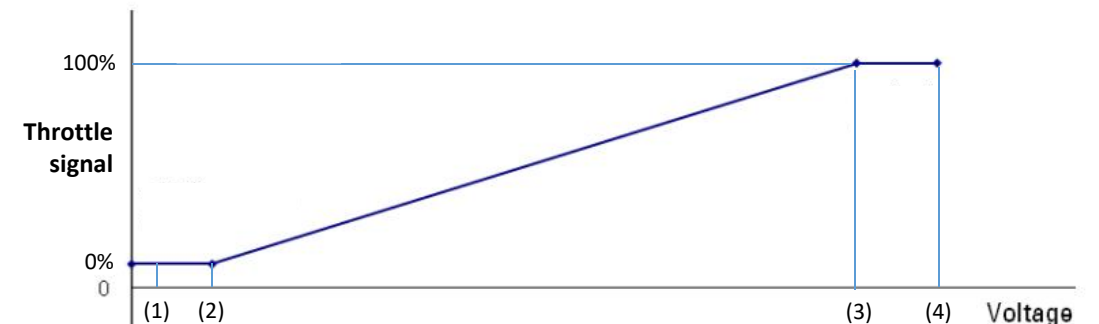
low protect

start voltage:

the end of the high

Parámetros del acelerador / Throttle Parameters

1. **Low protection value:** Si el valor de tension medido es inferior a ese valor, aparece un error en la señal del acelerador (valor muy bajo). If the measured voltage value is lower than this value, an error 95 appears in the throttle signal (value too low)
2. **Starting voltage:** Valor del voltaje a partir del cual el motor comienza a funcionar (sería el punto del 0% del acelerador). Voltage value from which the motor starts to work (it would be the 0% throttle point).
3. **End voltage:** Este valor marca el valor de tensión que establece el 100% del acelerador. Todo valor de tension superior, se seguirá considerando 100%. This value marks the voltage value that establishes 100% throttle. Any higher voltage value will continue to be considered 100% (The upper limit of the speed effective value, the max. speed value voltage.)
4. **High protection value:** Cuando el valor de tension sobrepasa este valor, aparece un error en la señal del acelerador (valor muy alto). When the voltage value exceeds this value, an error appears in the throttle signal (value too high).



VOTOL-EM-V3 Software Debugging

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SW: 0 HW: 0 buadrate: 115200 ☒ CAN enable

Basic Settings

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Overvoltage(V):

Undervoltage(V):

Soft undervoltage(V):

Undervoltage variation:

Busbar current(A):

Phase current:

throttle voltage set(max. 5.5v)

low protect:

start voltage:

the end of the high:

start setting

start torque:

combinative:

rate of rise:

rate of decline:

Start setting adjustment

- **Start torque:** Torque parameter at startup, usually set to 0 and not changed.
- **Combinative torque:** Starting delay parameter, the range is 0-350. The larger the value, the more obvious the effect of the starting delay.
- These two parameter above is effective for **middle motor with gear**
- **rate of rise :** Acceleration slope, the range is 10-255, the larger the value, the faster acceleration.
- **rate of decline:** Deceleration slope, 10-200 range value, the larger value, the more obvious the speed drop, and the inertia of the loosening handle is reduced.

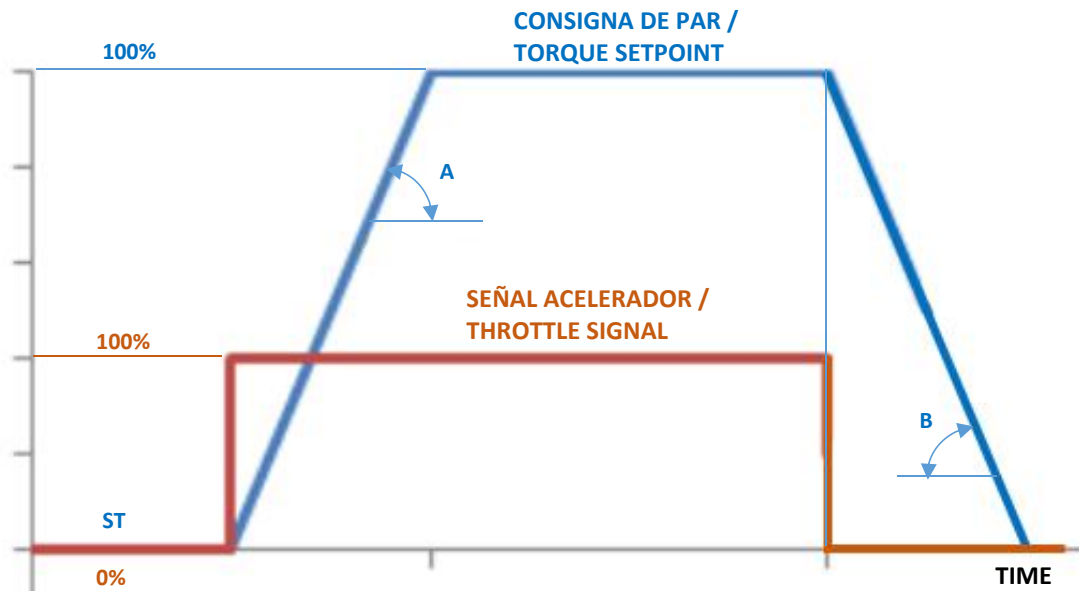
start setting

start torque:

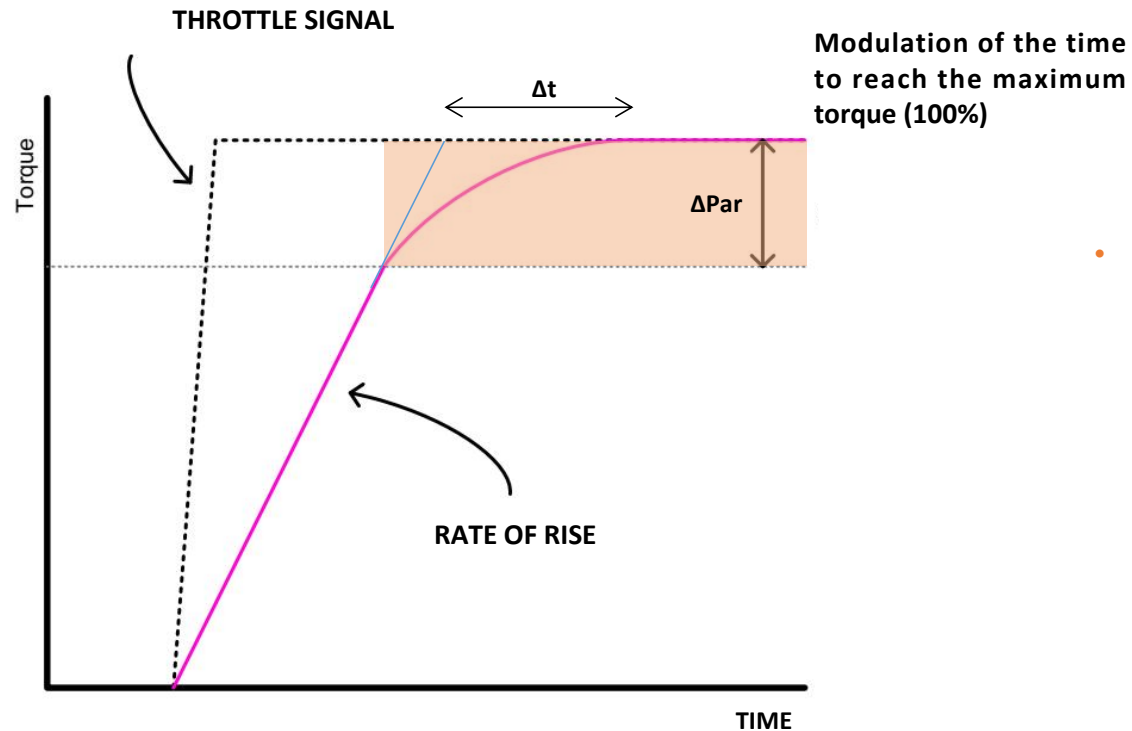
combinative:

rate of rise:

rate of decline:



- ST = Start Torque Rate of rise = $\text{tg}(A)$
- Rango de regulación / Regulation range = 10-250
- A mayor pendiente, más rápida es la respuesta (más rápido se alcanzará la consigna de par). The greater the slope, the faster the response (the faster the torque setpoint will be reached)
- Rate of decline = $\text{tg}(B)$
- Rango de regulación / Regulation range = 10-200
- A mayor pendiente, más rápida es la respuesta (más rápido se alcanzará la consigna de par). The greater the slope, the faster the response, the release speed of the handlebar drops faster (the faster the torque setpoint will be reached)



- **Combinative torque**
 - Rango de regulación / Regulation range = 0-350
 - As the value increases, more time is needed to reach the maximum torque setpoint. In this way, the behavior of the motor, in terms of acceleration, is smoothed out.
 - If the value 0 is specified, it means that there is no modulation, so the slope defined as Rate of Rise is followed until the torque setpoint is reached.

Current Limiting (A) = Valor de la corriente máxima permitida de alimentación al motor.
Value of the maximum current allowed to supply the motor

- Se debe especificar la corriente máxima admisible del motor eléctrico, que será la que aplique durante el modo S. The maximum allowable current of the electric motor during this driving mode S.
- El límite del controlador es de 150A (72V) (Pico de intensidad). **Controller limit is 150A (72V)**



Sport Mode = S Gear Tick this to activate Sport mode

Parameterization of driving mode Sport (S)

The motor is running in the flux weakening region

VOTOL-EM-V3 Software Debugging

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Sport mode setup

Current-Limiting(A): 0

Flux-Weakening Value: 0 0

☐ Automatic logout enadlers

Logout time(S): 0

Recovery time(S): 0

Downhill electric brake assist(HDC/HHC)

☐ HHC Enable

☐ HDC Enable

HDC lowrst speed: 0

Speed limit setting

☐ Speed limited enable

Speed ratio(%): 0

Flux-Weakening compensation: 0

Three-speed

Low(%): 0 0

Mid(%): 0 0

Hige(%): 0 0

Mid Flux-Weake 0 0

Hige Flux-Weake ning 0 0

Button/Switch 3 speed

☒ Button 3 speed ☐ Switch 3 speed

Three speed default gear

☒ Low ☐ Mid ☐ Hige

☐ Soft start enabled

Soft start grade: 1

Flux-Weakening Value:

- Flux weakening value at left side, range is 0-4000

When controller turn into S gears, the motor will be in flux weaken condition, speed increased. Setting value range is determined by motor condition:

Wheel hub motor, magnet high lower than 35mm	≤1500
Wheel hub motor, magnet high lower than 50mm	≤2300
Mid drive motor, surface attached magnetic steel	≤2300
Mid drive motor, V magnet steel	≤3000

Flux weakening adjustment at right side, range is 100-1200.
larger value, the motor more shake, smaller value, the motor less shake
Adjust the value in multiples of 50.

VOTOL-EM-V3 Software Debugging

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Sport mode setup

Current-Limiting(A):

Flux-Weakening Value:

☐ Automatic logout enadlers

Logout time(S):

Recovery time(S):

Downhill electric brake assist(HDC/HHC)

☐ HHC Enable

☐ HDC Enable

HDC lowrst speed:

Speed limit setting

☐ Speed limited enable

Speed ratio(%):

Flux-Weakening compensation:

Three-speed

Low(%):

Mid(%):

Hige(%):

Mid Flux-Weake

Hige Flux-Weake ning

Button/Switch 3 speed

☒ Button 3 speed ☐ Switch 3 speed

Three speed default gear

☒ Low ☐ Mid ☐ Hige

☐ Soft start enabled

Soft start grade:

VOTOL-EM-V3 Software Debugging

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Sport mode setup

Current-Limiting(A): 0

Flux-Weakening Value: 0 0

☐ Automatic logout enadlers

Logout time(S): 0

Recovery time(S): 0

Downhill electric brake assist(HDC/HHC)

☐ HHC Enable

☐ HDC Enable

HDC lowrst speed: 0

Speed limit setting

☐ Speed limited enable

Speed ratio(%): 0

Flux-Weakening compensation: 0

Three-speed

Low(%): 0 0

Mid(%): 0 0

Hige(%): 0 0

Mid Flux-Weake 0 0

Hige Flux-Weake ning 0 0

Button/Switch 3 speed

☒ Button 3 spee ☐ Switch 3 spee

Three speed default gear

☒ Low ☐ Mid ☐ Hige

☐ Soft start enabled

Soft start grade: 1

Flux weakening compensation factor

The hub motor is 10~20, this parameter can adjust the size of the reverse torque.

The mid-mounted motor is recommended to be 65~100, which can adjust the smoothness of reversing.

VOTOL-EM-V3 Software Debugging

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Sport mode setup

Current-Limiting(A): 0

Flux-Weakening Value: 0 0

☐ Automatic logout enadlers

Logout time(S): 0

Recovery time(S): 0

Downhill electric brake assist(HDC/HHC)

☐ HHC Enable

☐ HDC Enable

HDC lowrst speed: 0

Speed limit setting

☐ Speed limited enable

Speed ratio(%): 0

Flux-Weakening compensation: 0

Three-speed

Low(%): 0 0

Mid(%): 0 0

Hige(%): 0 0

Mid Flux-Weake 0 0

Hige Flux-Weake ning 0 0

Button/Switch 3 speed

☒ Button 3 spee ☐ Switch 3 spee

Three speed default gear

☒ Low ☐ Mid ☐ Hige

☐ Soft start enabled

Soft start grade: 1

Speed limit setting :
Choose if the speed limit or unlimit. The default is unlimited speed.

Automatic Logout Enablers

- Logout time: e.g. 30s after enter S gear sport mode (time is optional), auto-exit S gear sport mode.
- Recovery time: auto-exit S gear, recover from to S gear time. During recovery time (Invalid by press S gear).

☐ Automatic logout enablers

Logout time(S):

Recovery time(S):

VOTOL-EM-V3 Software Debugging

PAGE1 | PAGE2 | PAGE3 | Display | Port Settings

Sport mode setup

Current-Limiting(A):

Flux-Weakening Value:

☐ Automatic logout enablers

Logout time(S):

Recovery time(S):

Downhill electric brake assist(HDC/HHC)

☐ HHC Enable

☐ HDC Enable

HDC lowrst speed:

Speed limit setting

☐ Speed limited enable

Speed ratio(%):

Flux-Weakening compensation:

Three-speed

Low(%):

Mid(%):

Hige(%):

Mid Flux-Weake

Hige Flux-Weake ning

Button/Switch 3 speed

☒ Button 3 speed ☐ Switch 3 speed

Three speed default gear

☒ Low ☐ Mid ☐ Hige

☐ Soft start enabled

Soft start grade:

HHC = Hill Hold Control

Specifies whether hill assist is enabled

Operating mode not available

HDC lowest speed

It's a erratum in the program. It actually specifies the maximum speed of the motor.

VOTOL-EM-V3 Software Debugging

PAGE1 | **PAGE2** | PAGE3 | Display | Port Settings

Sport mode setup

Current-Limiting(A): 0

Flux-Weakening Value: 0 0

☐ Automatic logout enadlers

Logout time(S): 0

Recovery time(S): 0

Flux-Weakening compensation: 0

Three-speed

Low(%): 0 0

Mid(%): 0 0

Hige(%): 0 0

Mid Flux-Weake 0 0

Hige Flux-Weake ning 0 0

Button/Switch 3 speed

☒ Button 3 speed ☐ Switch 3 speed

Three speed default gear

☒ Low ☐ Mid ☐ Hige

☐ Soft start enabled

Soft start grade: 1

Downhill electric brake assist(HDC/HHC)

☐ HHC Enable

☐ HDC Enable

HDC lowrst speed: 0

Speed limit setting

☐ Speed limited enable

Speed ratio(%): 0

☒ HDC Enable

HDC lowrst speed: 840

Top Speed in rpm

HHC just suit for three wheeler, not works for two wheelers.
when HHC enabled, the vehicle will hold still when switching from the brake to throttle pedal.

VOTOL-EM-V3 Software Debugging

PAGE1 | PAGE2 | PAGE3 | Display | Port Settings

Sport mode setup

Current-Limiting(A): 0

Flux-Weakening Value: 0 0

☐ Automatic logout enadlers

Logout time(S): 0

Recovery time(S): 0

Downhill electric brake assist(HDC/HHC)

☐ HHC Enable

☐ HDC Enable

HDC lowrst speed: 0

Speed limit setting

☐ Speed limited enable

Speed ratio(%): 0

Flux-Weakening compensation: 0

Three-speed

Low(%): 0 0

Mid(%): 0 0

Hige(%): 0 0

Mid Flux-Weake 0 0

Hige Flux-Weake ning 0 0

Button/Switch 3 speed

☒ Button 3 speed ☐ Switch 3 spee

Three speed default gear

☒ Low ☐ Mid ☐ Hige

☐ Soft start enabled

Soft start grade: 1

Configuración de los modos normales de funcionamiento.
Configuration of the normal modes of operation

3 speed setting

Low(%): 60

Mid(%): 80

Hige(%): 100

3 current limit setting

100

100

7

Low speed = $60\% \times \text{HDC lowest speed}$
Mid speed = $80\% \times \text{HDC lowest speed}$
High speed = $100\% \times \text{HDC lowest speed}$

4. Overshoot rpm parameter. It will increase the rated rpm by increase this RPM. 0~14 from hub motor and 14 for IPM motor. It may affect the efficiency and performance of the power train by changing this value. Not suggest to be adjusted



VOTOL-EM-V3 Software Debugging

PAGE1 | PAGE2 | PAGE3 | Display | Port Settings

Sport mode setup

Current-Limiting(A): 0

Flux-Weakening Value: 0 0

☐ Automatic logout enadlers

Logout time(S): 0

Recovery time(S): 0

Downhill electric brake assist(HDC/HHC)

☐ HHC Enable

☐ HDC Enable

HDC lowrst speed: 0

Speed limit setting

☐ Speed limited enable

Speed ratio(%): 0

Flux-Weakening compensation: 0

Three-speed

Low(%): 0 0

Mid(%): 0 0

Hige(%): 0 0

Mid Flux-Weake 0 0

Hige Flux-Weake 0 0

Flux-Weake ning

Button/Switch 3 speed

☒ Button 3 speed ☐ Switch 3 speed

Three speed default gear

☒ Low ☐ Mid ☐ Hige

☐ Soft start enabled

Soft start grade: 1

此处值不可更改 The value can not be changed.

Mid Flux-Weake ning Value: 4000 0

Hige Flux-Weake ning Value: 8000 6000



Escoger el tipo de mando que establece los 3 modos de funcionamiento:

- Button = Botón mediante el cual se establece el modo en función del número de veces que se presione
- Switch = Interruptor de 3 posiciones

Choose the type of control that establishes the 3 operating modes:

Button = Button through which the mode is set based on the number of times it is pressed

Switch = 3 position switch

VOTOL-EM-V3 Software Debugging

PAGE1 | PAGE2 | PAGE3 | Display | Port Settings

Sport mode setup

Current-Limiting(A):

Flux-Weakening Value:

☐ Automatic logout enadlers

Logout time(S):

Recovery time(S):

Downhill electric brake assist(HDC/HHC)

☐ HHC Enable

☐ HDC Enable

HDC lowrst speed:

Speed limit setting

☐ Speed limited enable

Speed ratio(%):

Flux-Weakening compensation:

Three-speed

Low(%):

Mid(%):

Hige(%):

Mid Flux-Weake

Hige Flux-Weake ning

Button/Switch 3 speed

☒ Button 3 speed ☐ Switch 3 speed

Three speed default gear

☒ Low

☐ Mid

☐ Hige

☒ Soft start enabled

Soft start grade:

Soft start selection, not hard start, the higher the level, the more obvious the starting power.

d:

g enable

:

Three speed default gear

☐ Low

☒ Mid

☐ Hige

☒ Soft start enabled

Soft start grade:

Optional

1. It can be checked. The larger the level is, the more obvious the starting power is
2. Uncheck for hard start

En el caso de escoger "Button 3 speed", establece la marcha que se activa por defecto.
In the case of choosing "Button 3 speed", it establishes the gear that is activated by default.
Choose the default gear

Nº de pares polos del motor / Number of motor pole pairs

El motor Varchea posee 30 pares de polos

The Varchea motor has 30 pairs of poles.

Different motor may have different pole pairs.

Motor wire sequence adjustment,
Hall wire sequence adjustment.



Motor rotor magnet fixing type,
the hall position in-hub motor

VOTOL-EM-V3 Software Debugging

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Motor Setting

Pole pairs:

☐ exchange hall wire color Yellow-Blue

☐ exchange phase wire color Blue-Green

Motor type

☒ surface-mount ☐ V-type

Hall shift Angle

Out-put

☒ One-Lin ☐ Hall speedometer

☐ Moving vehicle booster ☐ Cruise

Moving vehicle booster Speed:

Moving vehicle booster torque:

Double voltage automatic identification setting

☐ Double-voltage ☒ Low ☐ Hige

Reversing the speed limit(%):

EBS ratio(%):

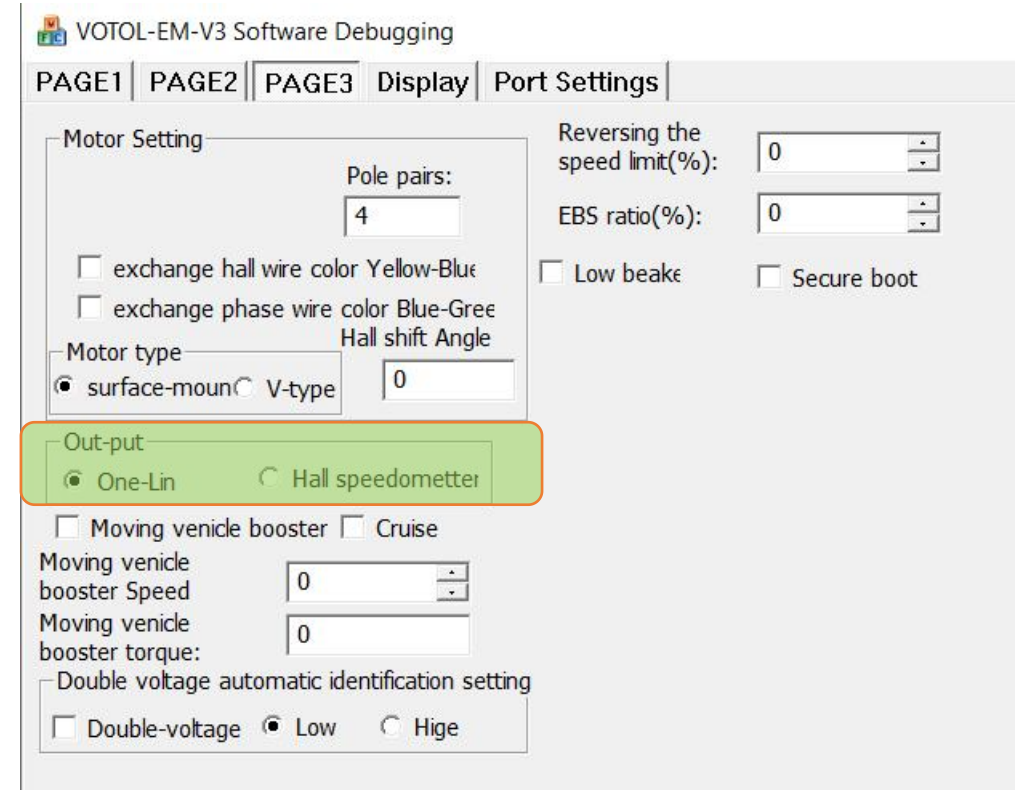
☐ Low beake ☐ Secure boot

Phase shift angle adjustment angle, in hub motor usually with -60.

Speedometer communication selection: ISDN or Hall signal

The controller to speedometer data output has 2 types: LIN protocol speedometer and hall speedometer, it needs to be decided by the vehicle's speedometer. We provide standard LIN protocol speedometer.

If we use CAN protocol, then we need to integrate with CAN speedometer.



VOTOL-EM-V3 Software Debugging

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Motor Setting

Pole pairs: 4

☐ exchange hall wire color Yellow-Blue

☐ exchange phase wire color Blue-Green

Motor type

☒ surface-mount ☐ V-type

Hall shift Angle: 0

Reversing the speed limit(%): 0

EBS ratio(%): 0

☐ Low beak ☐ Secure boot

Out-put

☒ One-Lin ☐ Hall speedometer

☐ Moving vehicle booster ☐ Cruise

Moving vehicle booster Speed: 0

Moving vehicle booster torque: 0

Double voltage automatic identification setting

☐ Double-voltage ☒ Low ☐ Hige

(1) Move assist function, valid after check

The speed of the transfer assist is selected as the percentage of the motor base speed. The default is 10%.

Moving assist torque value 320, corresponds to the torque value is 9~10N.m (varies depending on the motor characteristics)

☐ Moving vehicle booster ☐ Cruise

Moving vehicle booster Speed: 10

Moving vehicle booster torque: 1000

Es un valor en porcentaje con respecto a la motor base speed.
It is a value in percentage with respect to the motor base speed.

The vehicle moving assist function is used for two-wheeled vehicles, the speed is less than 3KM/H, and the torque is 9-19N.M.

(2) Cruise Function, valid after check

Turn to a certain angle and maintain more than 8 seconds, enter into the cruise control mode, any operation to exit the cruise mode (make a brake, turn the handle)

VOTOL-EM-V3 Software Debugging

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Motor Setting

Pole pairs: 4

☐ exchange hall wire color Yellow-Blue

☐ exchange phase wire color Blue-Gree

Motor type: ☒ surface-moun ☐ V-type

Hall shift Angle: 0

Out-put: ☒ One-Lin ☐ Hall speedometer

☐ Moving vehicle booster ☐ Cruise

Moving vehicle booster Speed: 0

Moving vehicle booster torque: 0

Double voltage automatic identification setting

☐ Double-voltage ☒ Low ☐ Hige

Reversing the speed limit(%): 0

EBS ratio(%): 0

☐ Low beake ☐ Secure boot

VOTOL-EM-V3 Software Debugging

PAGE1 | PAGE2 | **PAGE3** | Display | Port Settings

Motor Setting

Pole pairs: 4

☐ exchange hall wire color Yellow-Blue

☐ exchange phase wire color Blue-Green

Motor type

☒ surface-mount ☐ V-type

Hall shift Angle: 0

Out-put

☒ One-Lin ☐ Hall speedometer

☐ Moving vehicle booster ☐ Cruise

Moving vehicle booster Speed: 0

Moving vehicle booster torque: 0

Double voltage automatic identification setting

☐ Double-voltage ☒ Low ☐ Hige

Reversing the speed limit(%): 0

EBS ratio(%): 0

☐ Low beake ☐ Secure boot

Límite máximo velocidad marcha atrás. Maximum reverse speed limit. Representa un porcentaje de las rpm máximas del motor. Represents a percentage of the maximum engine rpm. Recomendación QS: No exceder el 30% (accidentes). QS recommendation: Do not exceed 30% (accidents)

EBS = Electronic Brake System

Establece el porcentaje de recuperación de energía en una frenada regenerativa Sets the percentage of energy recovery in regenerative braking, two wheeler with 15-20, while three or four wheeler with 20-30.

Safety Switch Function valid checked

When using the safety switch function, need to select the safety switch (Kindly note the vehicle is running)

Low brake enable valid check

VOTOL-EM-V3 Software Debugging

PAGE1 | PAGE2 | **PAGE3** | Display | Port Settings

Motor Setting

Pole pairs: 4

☐ exchange hall wire color Yellow-Blue

☐ exchange phase wire color Blue-Green

Motor type: ☒ surface-mount ☐ V-type

Hall shift Angle: 0

Reversing the speed limit(%): 0

EBS ratio(%): 0

☐ Low beake ☐ Secure boot

Out-put: ☒ One-Lin ☐ Hall speedometer

☐ Moving venicle booster ☐ Cruise

Moving venicle booster Speed: 0

Moving venicle booster torque: 0

Double voltage automatic identification setting

☐ Double-voltage ☒ Low ☐ Hige

Automatically switch dual voltage mode

mode	Volt for switching to low volt mode	Volt for switching to high volt mode
48~60V	<49V	>63.5V
60~72V	<61V	>77V
72~84V	<72V	>93.5V

ECU available on 60V platform and 72V platform

3.5 Dual voltage setting: default single voltage.

Note: The dual voltage function speed parameter value is followed by the motor speed setting. Adjust the parameters on the setup page 2.

Dual voltage setting, default single volage, low voltage

VOTOL-EM-V3 Software Debugging

PAGE1 | PAGE2 | PAGE3 | Display | Port Settings

Motor Setting

Pole pairs:

☐ exchange hall wire color Yellow-Blue
☐ exchange phase wire color Blue-Green

Motor type
☒ surface-mount ☐ V-type

Out-put
☒ One-Lin ☐ Hall speedometer

☐ Moving vehicle booster ☐ Cruise

Moving vehicle booster Speed:
Moving vehicle booster torque:

Double voltage automatic identification setting
☐ Double-voltage ☒ Low ☐ Hige

Reversing the speed limit(%):
EBS ratio(%):
☐ Low beake ☐ Secure boot

PORT: COM3 OPEN connect success_count: 0
status: research param write error_count: 0
import prarm save prarm



communication and download set

set compare check
current check

remote control

Throttle: 0.50 V

Gear
☒ L ☐ M ☐ H ☐ S ☐ Switch

☐ Brake ☐ R ☐ Lock

Calibration

vol cal: CLR DTC
cur cal:
weak flux:

open file
download

moden: No-ops delay: 12

comunicatin and download set

set compare check

current check

remote control

Throttle: 0.50 V

Gear

☒ L ☐ M ☐ H ☐ S

☐ Switch

☐ Brake ☐ R ☐ Lock

Calibration

vol cal: 0

cur cal: 0

weak flux: 3435973

open file

download

moden: No-ops delay: 12

Checking pupose, no need to fill in.

Analog handle voltage.

Low speed, Analog handle voltage, high speed, sport speed

☐ switch

Switch: if no select, display speed/voltage/current; If select, display speed/Q-axis voltage/D-axis voltage.

☐ Brake ☐ R ☐ Lock

Brake/Reverse/Lock

Open the ECU program file (bin file) or configuration file (ini file)

Download: Import the ECU program file or configuration file into the controller.

communication and download set

set compare check

current check

remote control

Throttle: 0.50 V

Gear

☒ L ☐ M ☐ H ☐ S ☐ Switch

☐ Brake ☐ R ☐ Lock

Calibration

vol cal: 0

cur cal: 0

weak flux: 3435973

open file

download

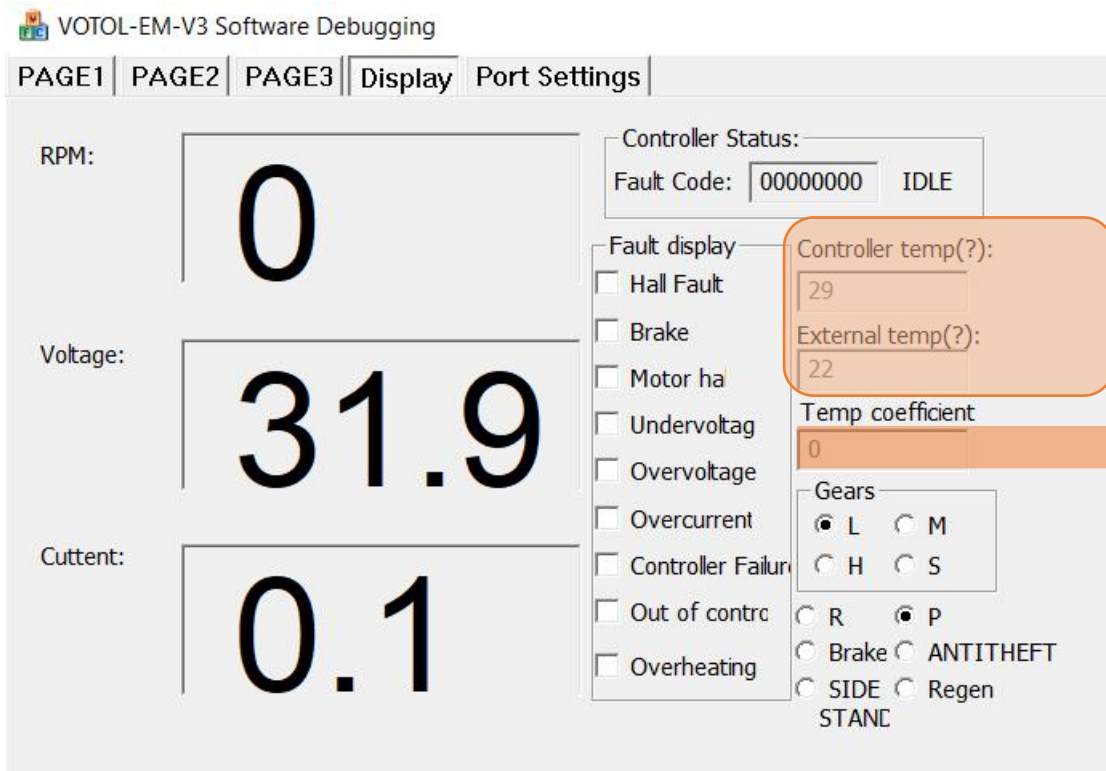
moden: No-ops delay: 12

Remove failure

Calibration values:

Voltage calibration: adjust whether the displayed voltage is consistent with the actual voltage, no need to change

Current calibration: adjust whether the displayed current is consistent with the actual current, no need to change
Weake flux value: adjust D-axis voltage and Q-axis current ,without changing



For V1 motor:

- When temperature is over 120°C, the controller shut down.
- When temperature less than 100°C, the controller work again.

Display the controller temperature

Display the motor temperature

Temperature Coefficient: Displays the controller hardware coefficient

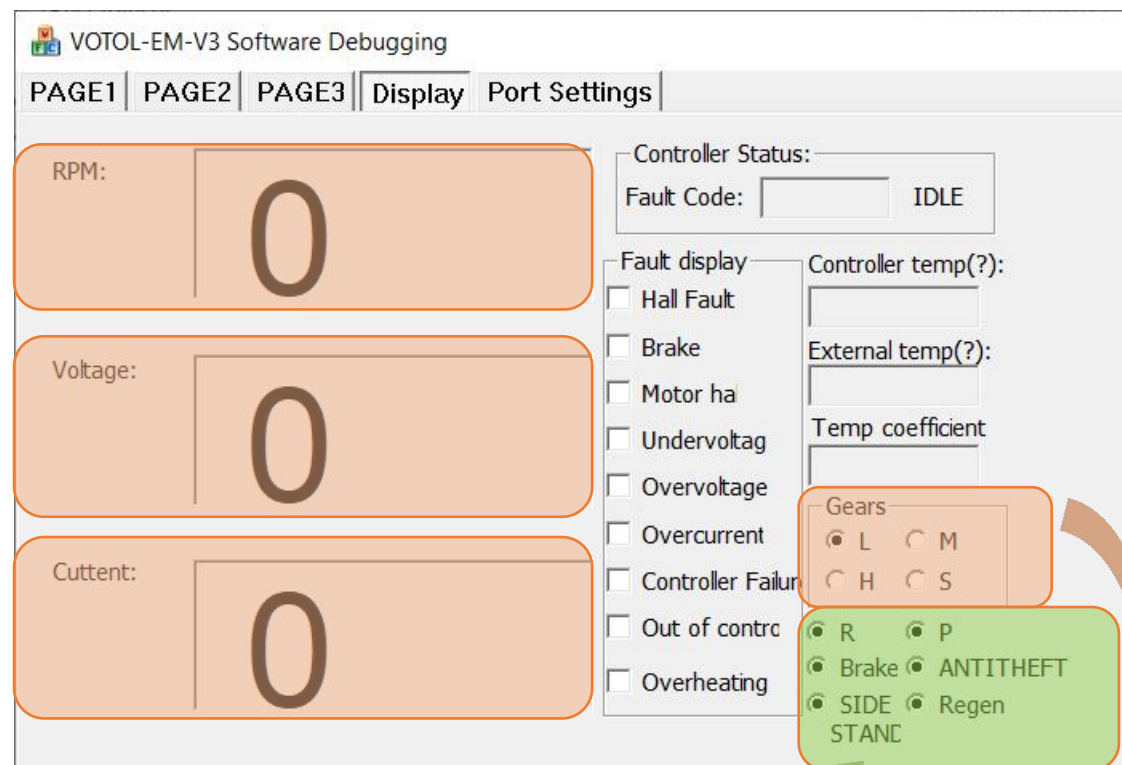
Shows the value of the rotation rpm of the motor rotor in real time

Shows the value of the supply voltage to the controller in the real time

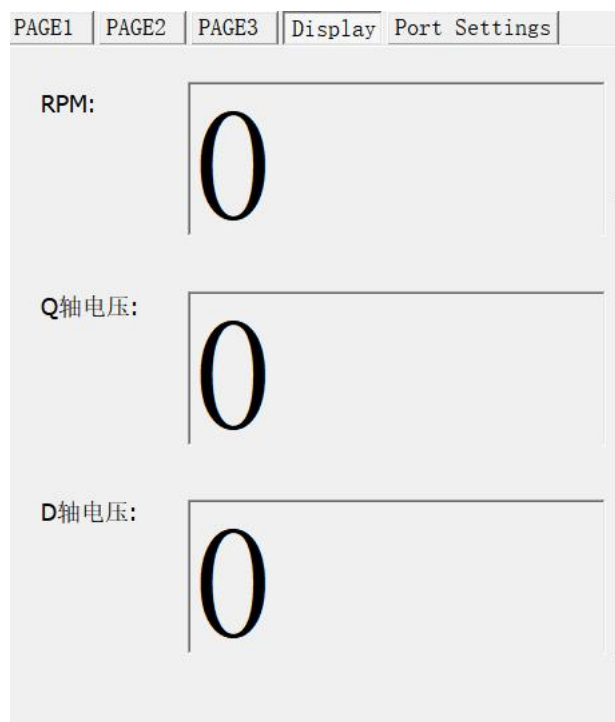
Show the current value of the supply to the controller in real time
(bus current value)

Show the active functions on the motor:

- R = Reversa / Reverse
- B = Frenando / Brake
- Side Stanc = Modo Sport / Sport Mode
- P = Posición Parking / Parking Position
- ANTI THEFT
- R = Regeneración / Regeneration



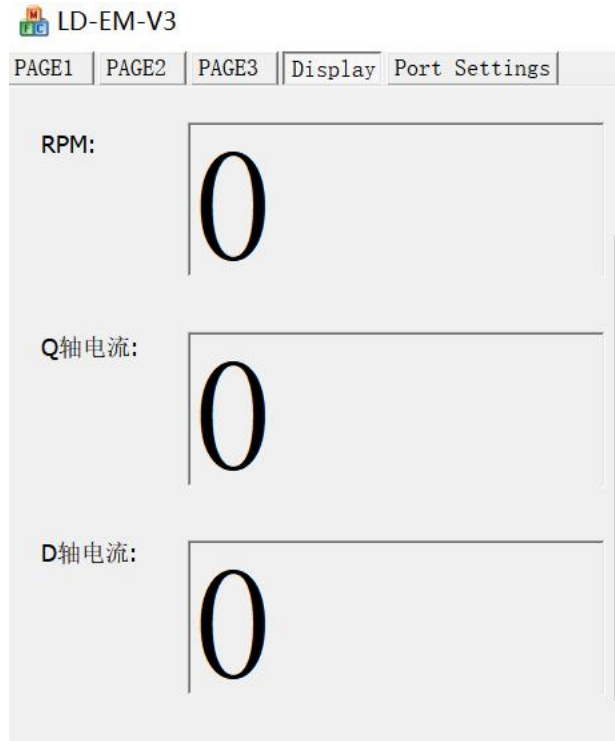
Shows the current gear mode



Q-axis voltage: the range is 2000-3000 when the motor angle is correct

D-axis voltage: the range is -600 to -1000 when the motor angle is correct (may fluctuate slightly)

After the weakening flux is adjusted to the final speed, adjust the D-axis voltage by adjusting the value of the weakening flux coefficient, the normal range is 0 to -300.



Q-axis current: Phase current limit corresponds to page 1.

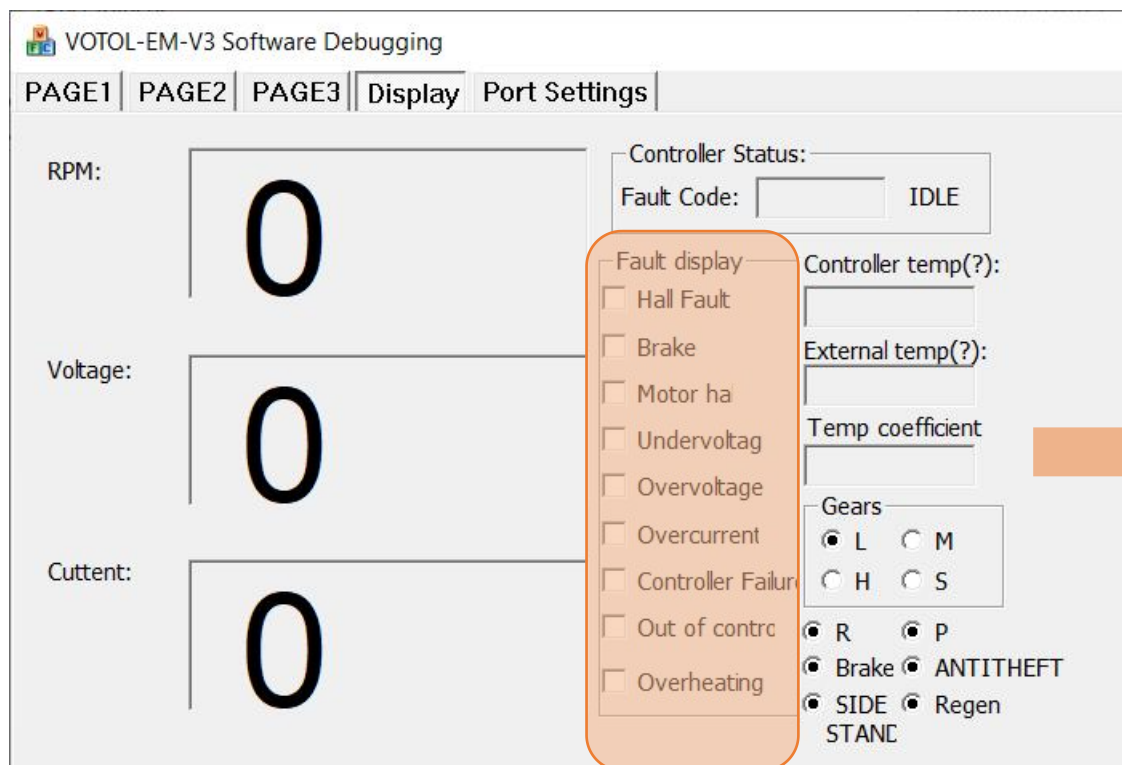
After the weakening flux is adjusted to the final speed, adjust the Q-axis current i current by adjusting the value of the weakening flux coefficient, the normal range is 100 to 200.

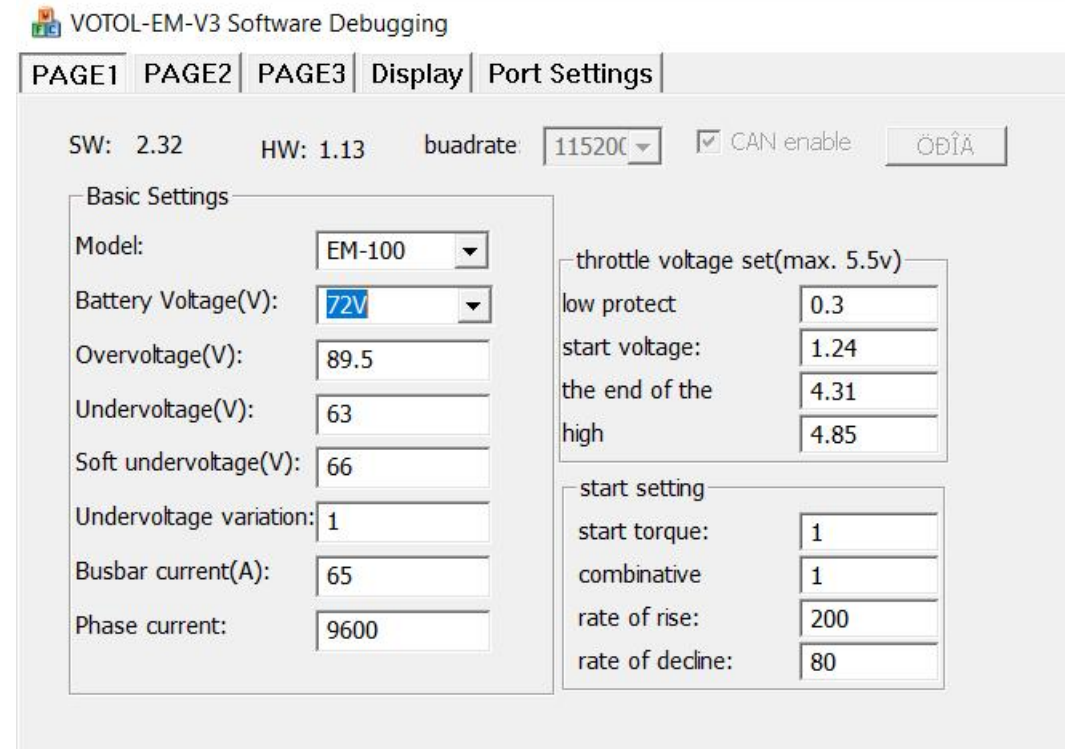
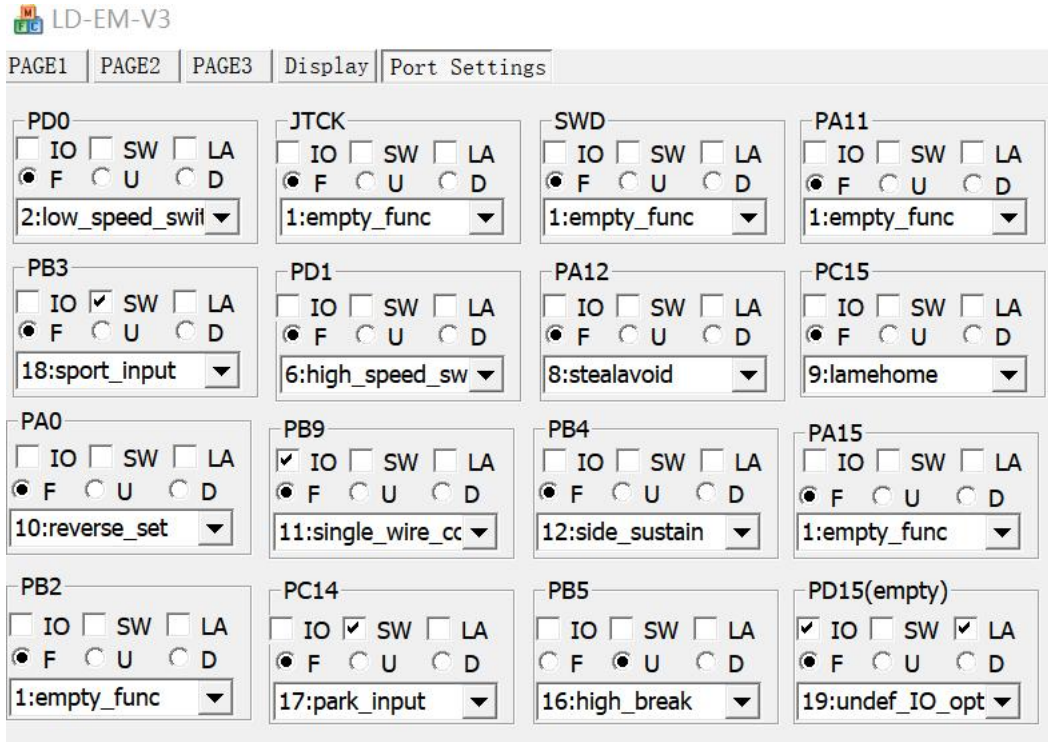
D-axis current: corresponds to the desired weakening flux value at the current speed.

If it is close to 0, weakening flux is not required for the current speed.

If with 500, the current speed needs 500 weakening flux to reach.

NOTE: The weakening flux value corresponds to the weakening flux value for the corresponding gear on page 2 of settings.

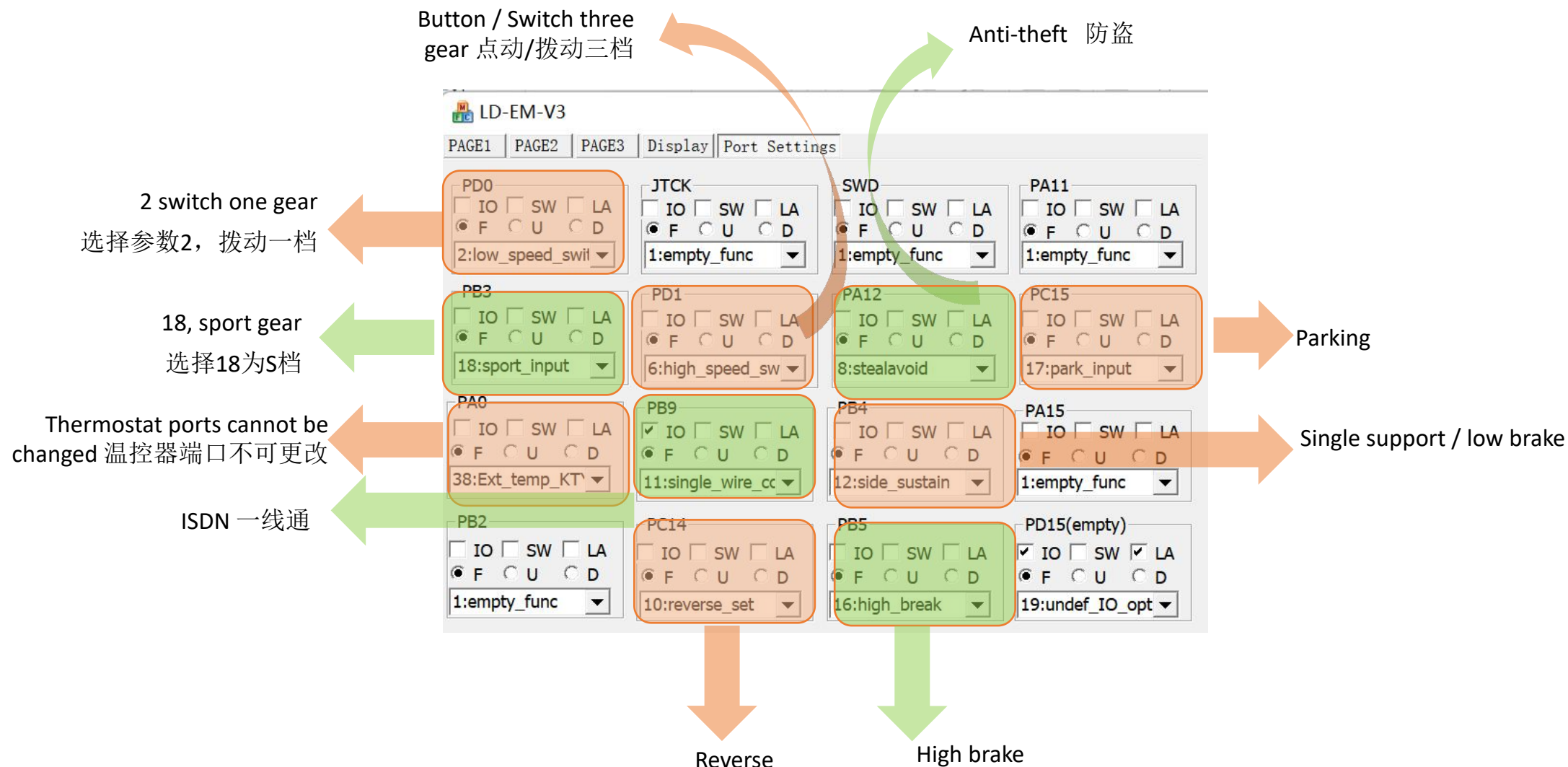


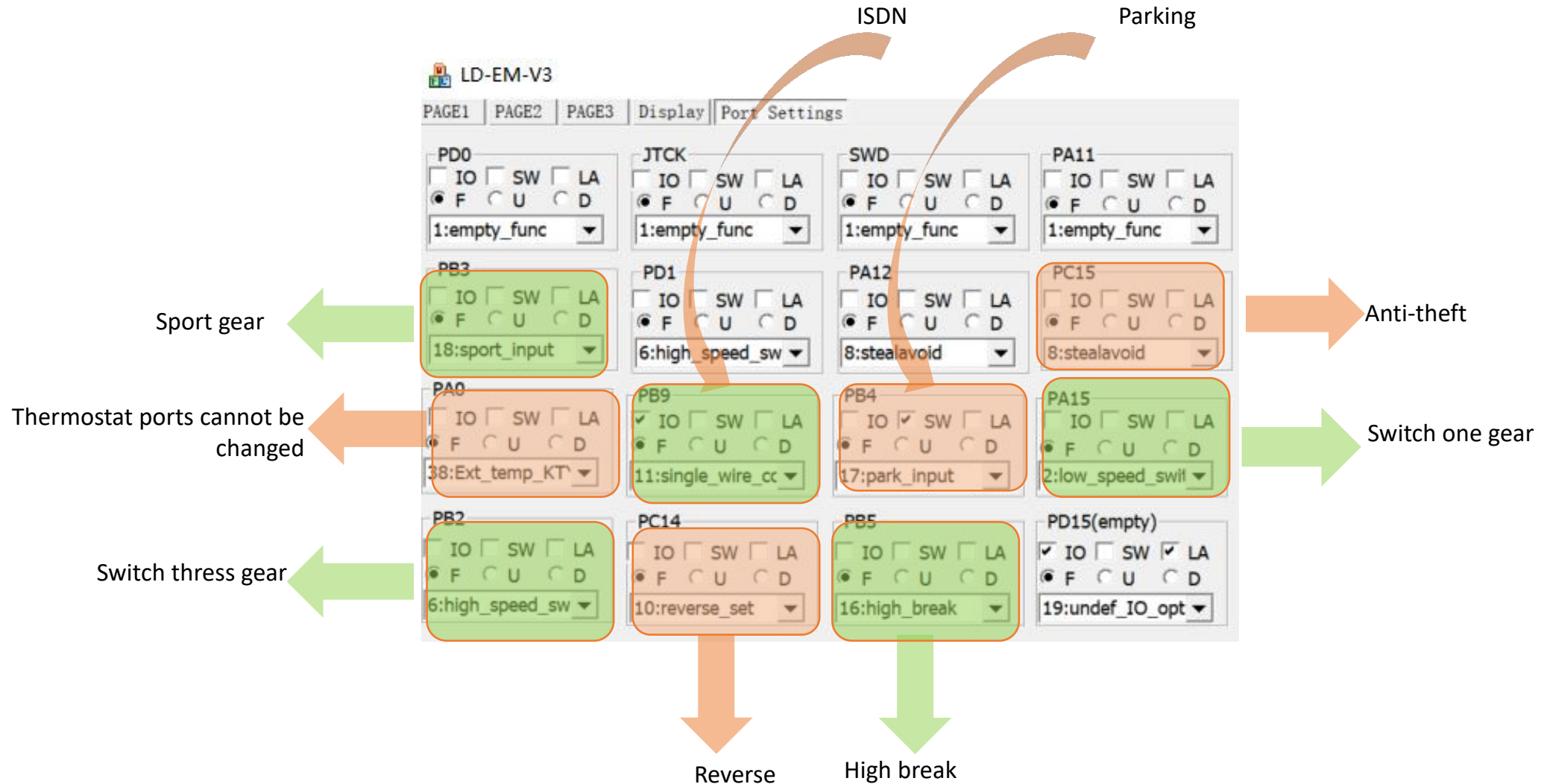


The above port functions are factory default configurations

PORT SETTING INTERFACE

EM30S EM50 (EM100 no anti-theft) -EM150S (before date 20200320) in common use

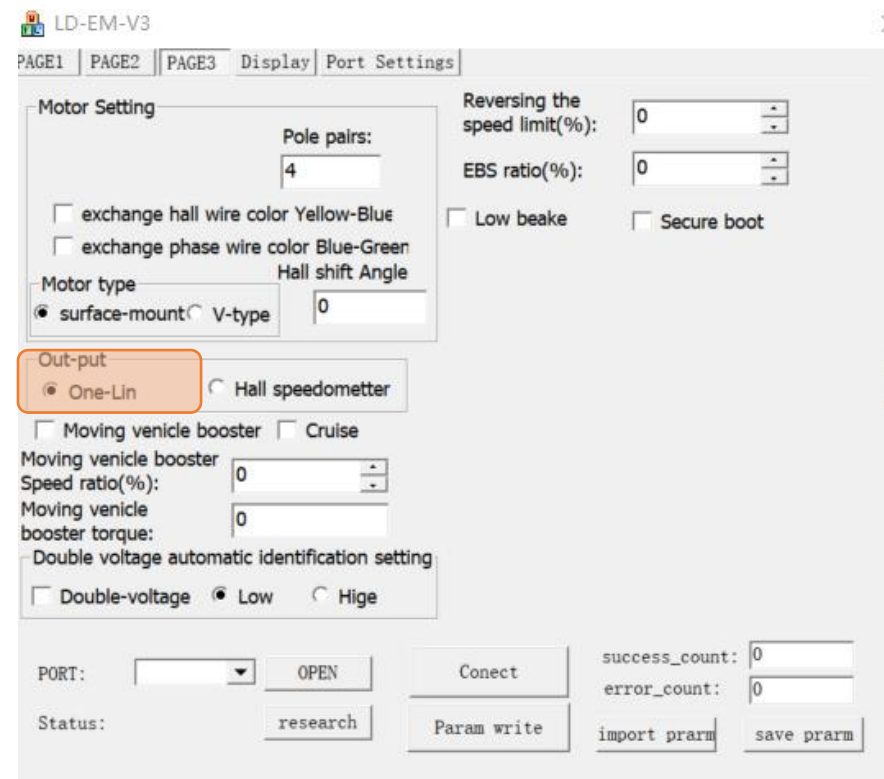
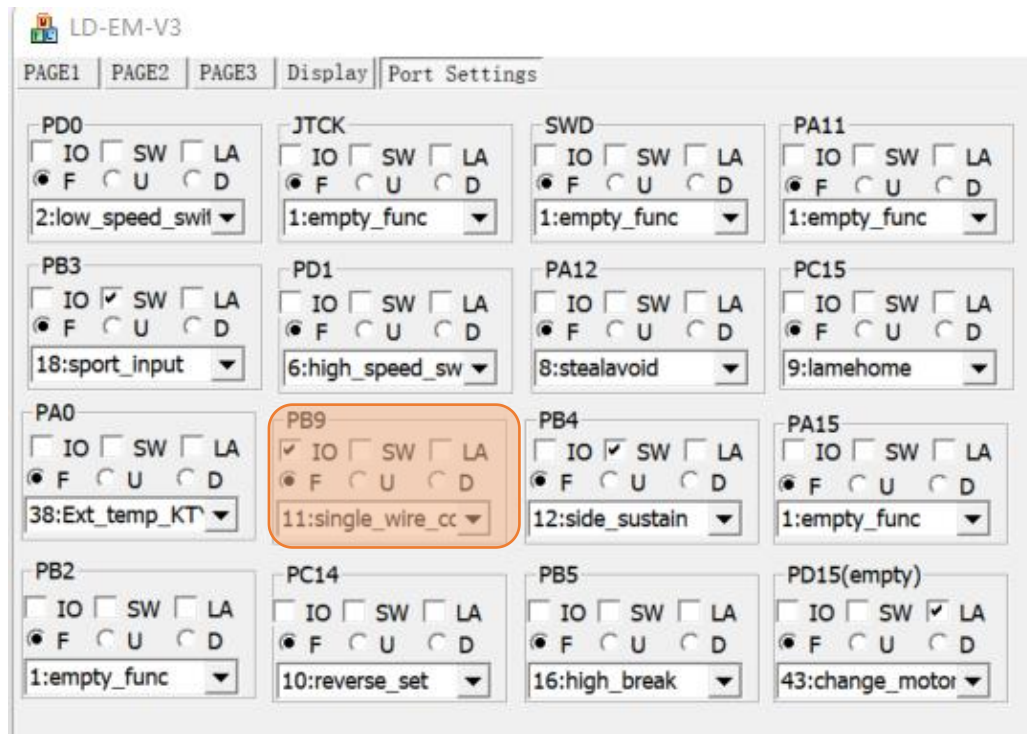


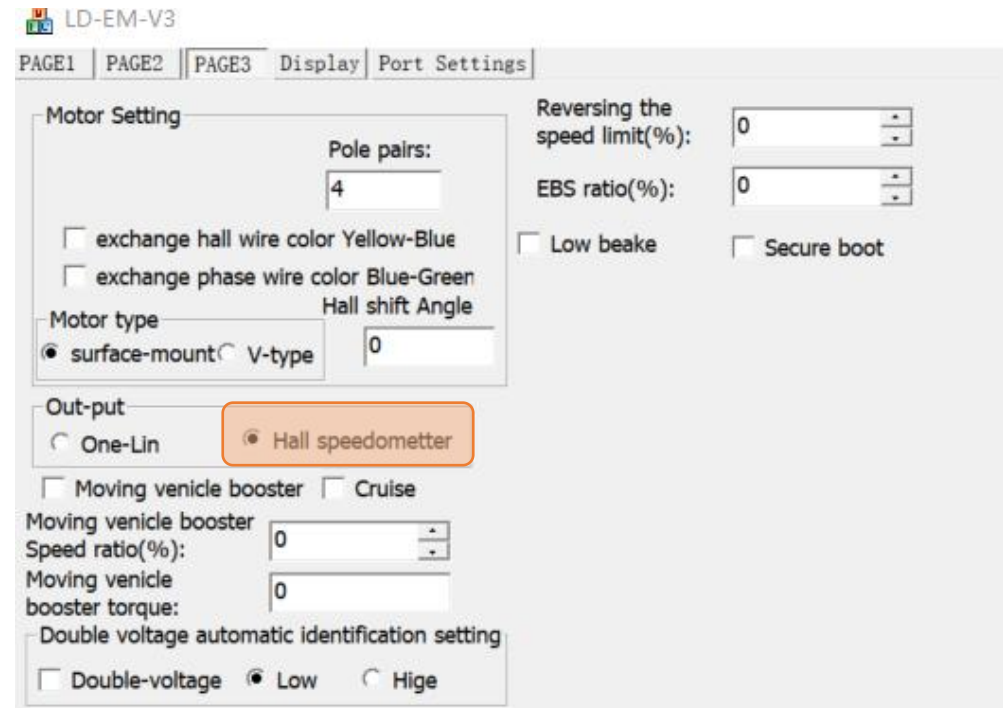
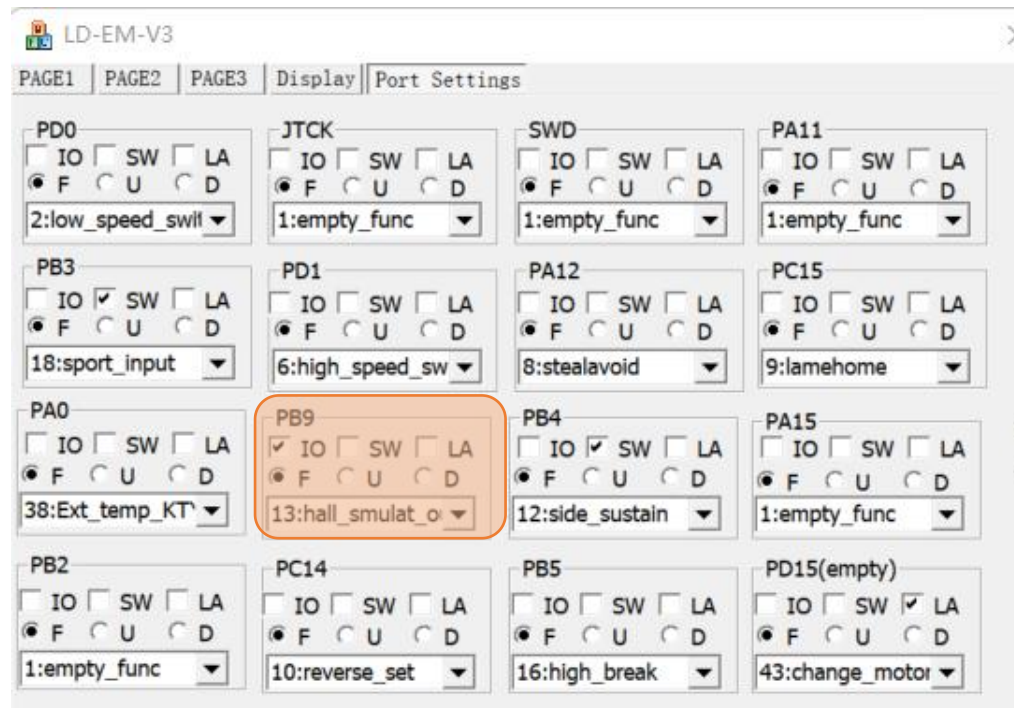


LD-EM-V3

PAGE1 | PAGE2 | PAGE3 | Display | Port Settings

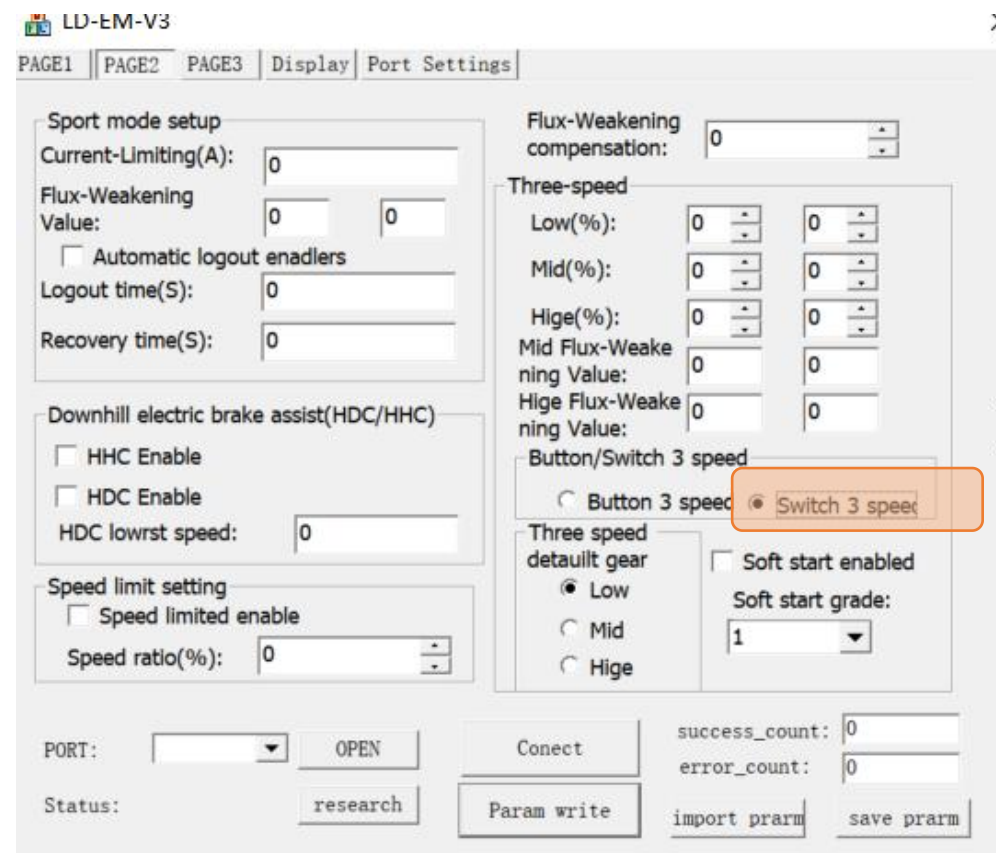
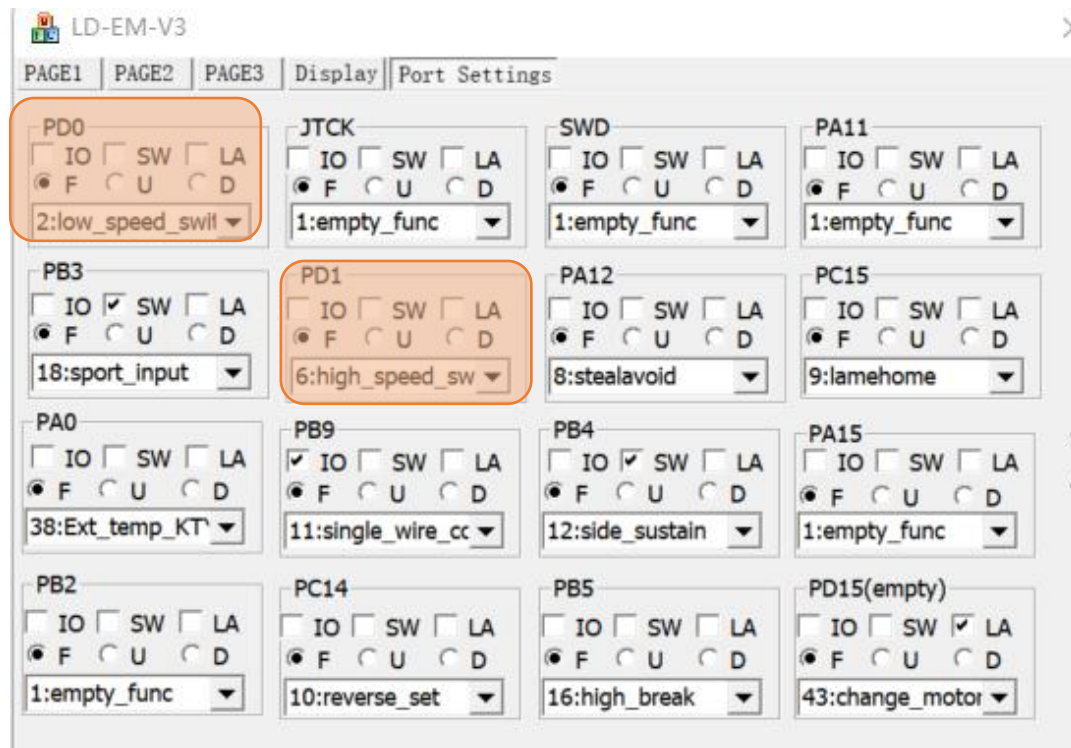
PD0 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 12:side_sustain ▼	JTCK ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func ▼	SWD ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func ▼	PA11 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func ▼
PB3 ☐ IO ☒ SW ☐ LA ☒ F ☐ U ☐ D 18:sport_input ▼	PD1 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 6:high_speed_sw ▼	PA12 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 8:stealavoid ▼	PC15 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 17:park_input ▼
PA0 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 38:Ext_temp_KT` ▼	PB9 ☒ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 11:single_wire_cc ▼	PB4 ☐ IO ☒ SW ☐ LA ☒ F ☐ U ☐ D 6:high_speed_sw ▼	PA15 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func ▼
PB2 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 2:low_speed_swit ▼	PC14 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 10:reverse_set ▼	PB5 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 16:high_break ▼	PD15(empty) ☒ IO ☐ SW ☒ LA ☒ F ☐ U ☐ D 19:undef_IO_opt ▼





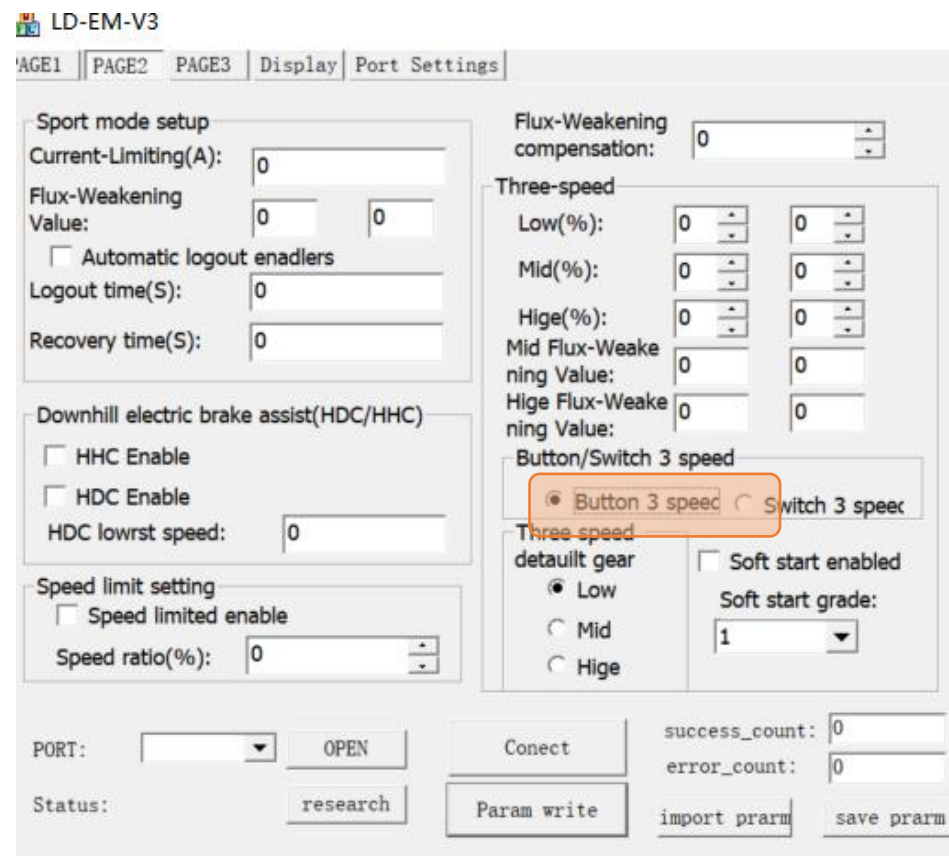
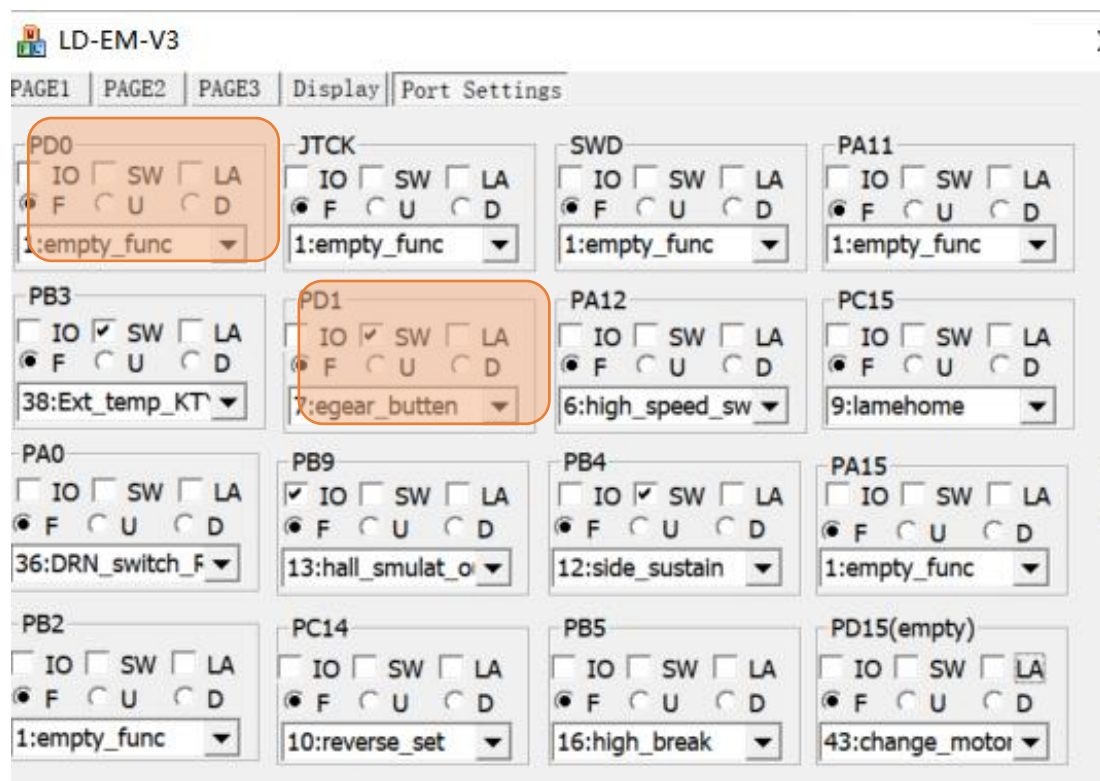
SWITCH THREE SPEED SETTING 拨动三速设置

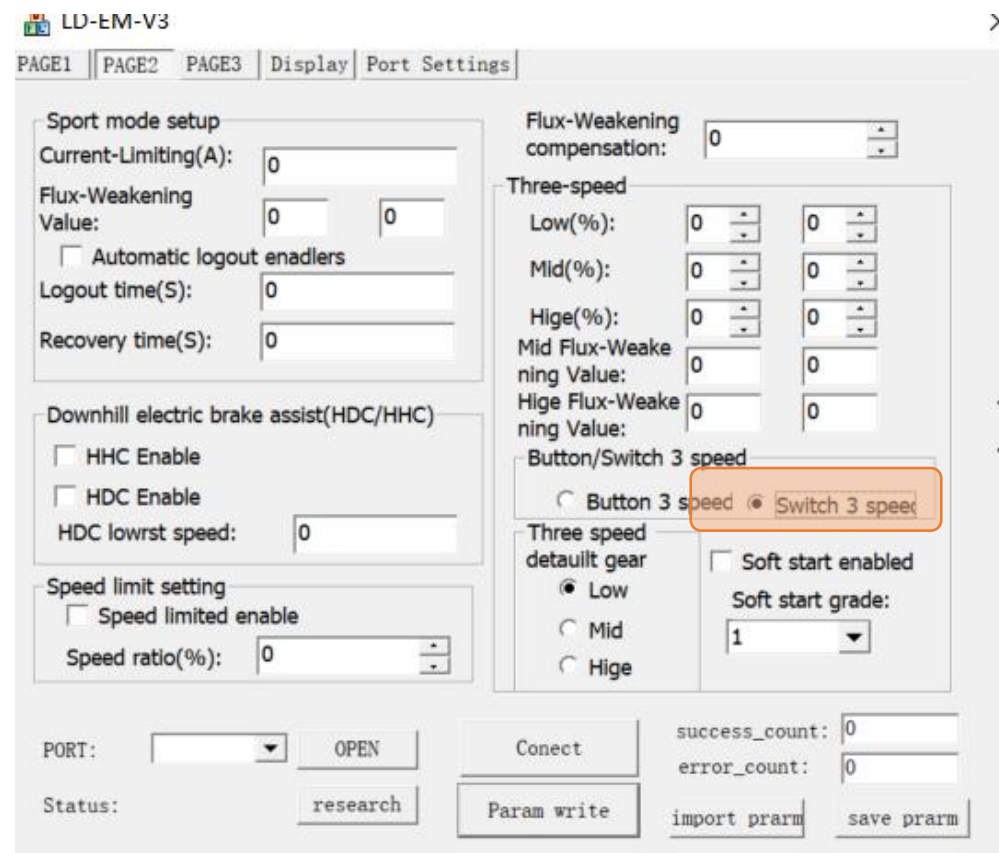
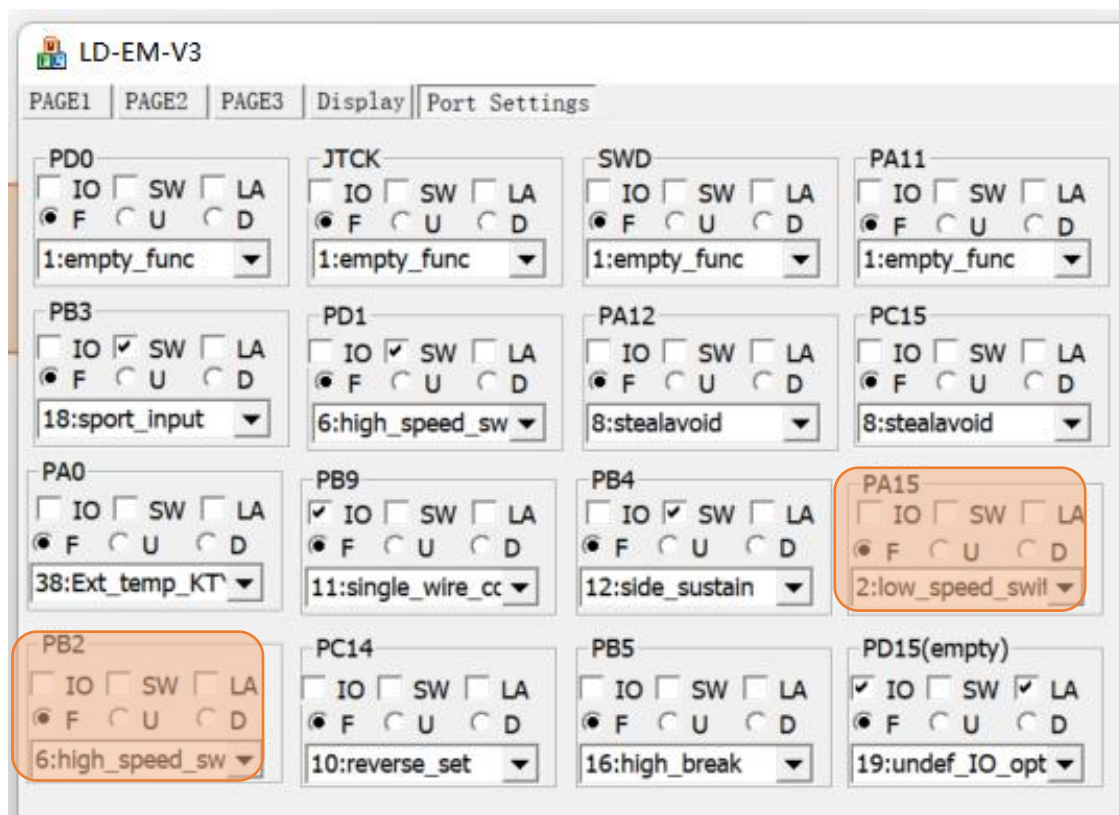
EM30S EM50 (EM100 no anti-theft) -EM150S (before date 20200320) in common use

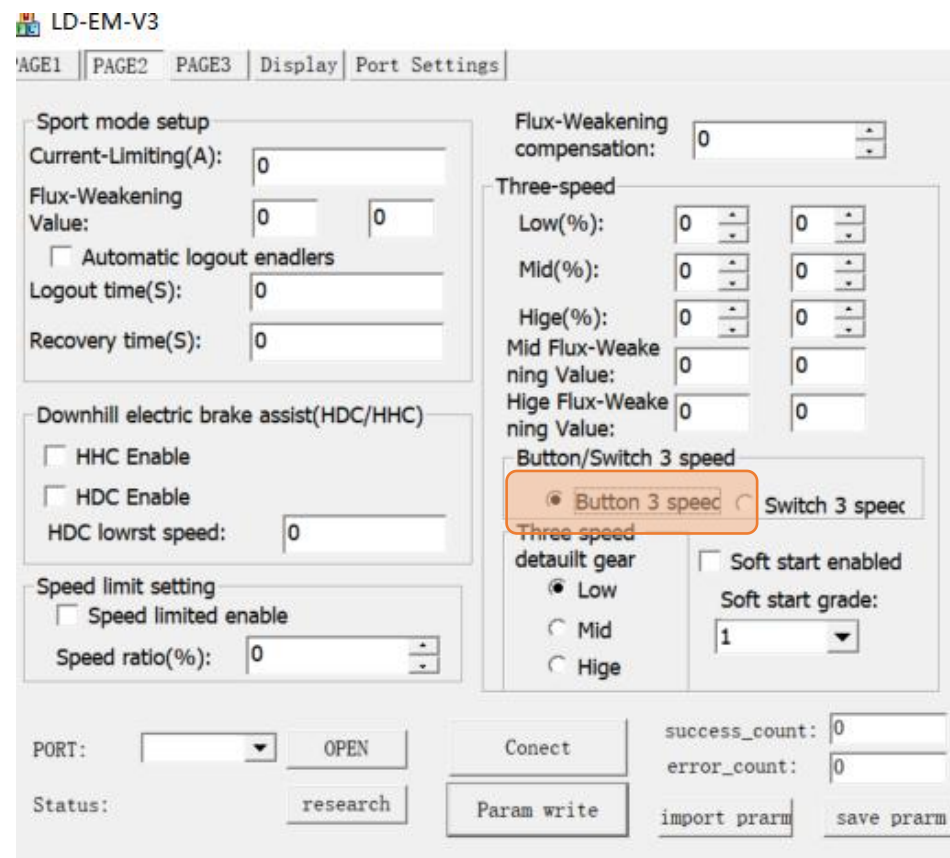
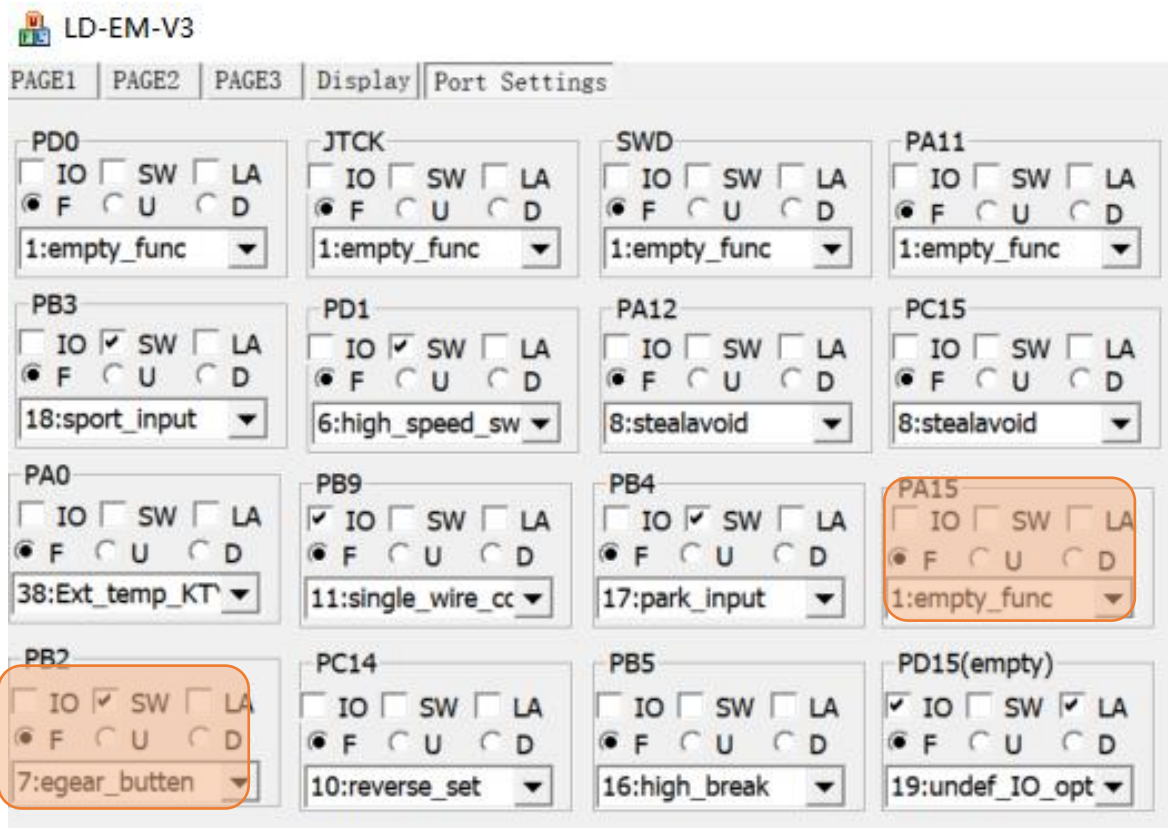


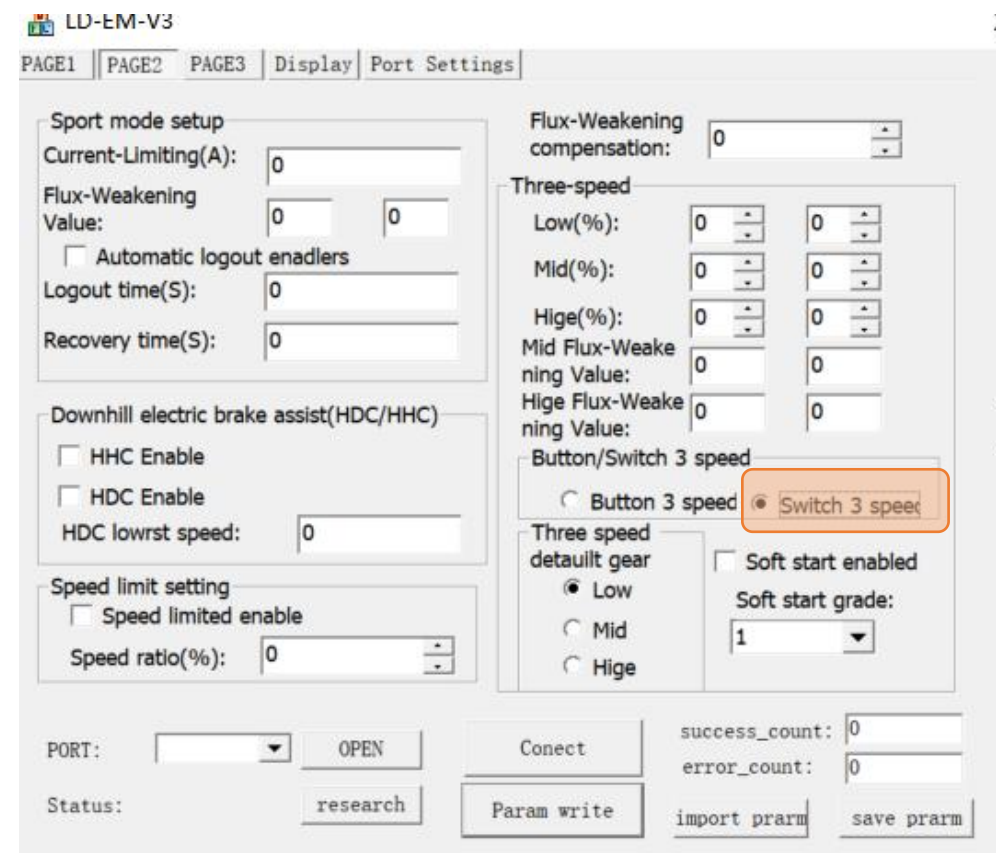
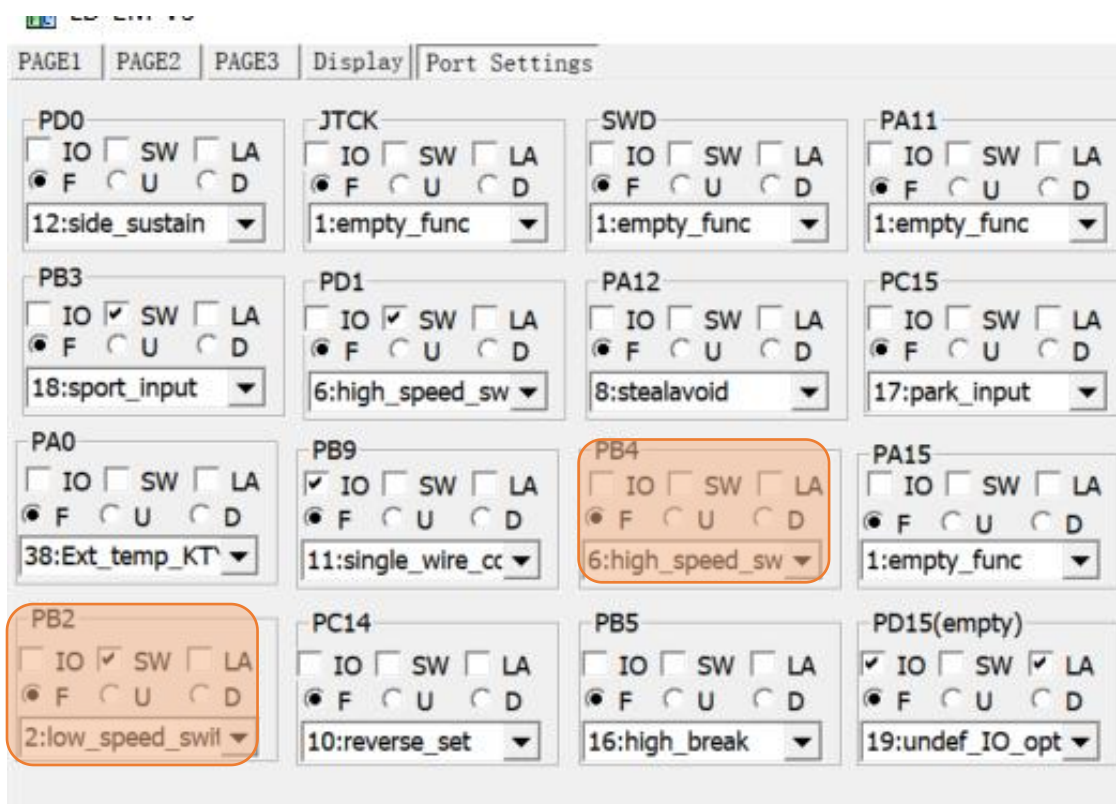
BUTTON THREE SPEED SETTING 点动三速设置

EM30S EM50 (EM100 no anti-theft) -EM150S (before date 20200320) in common use





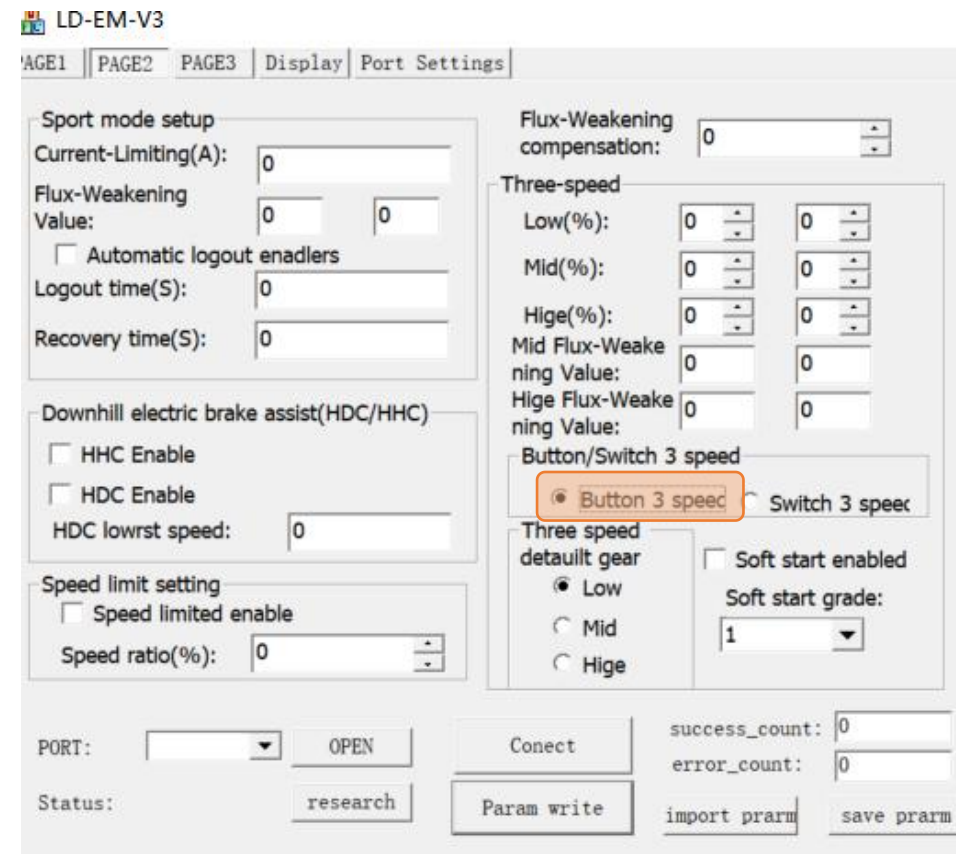
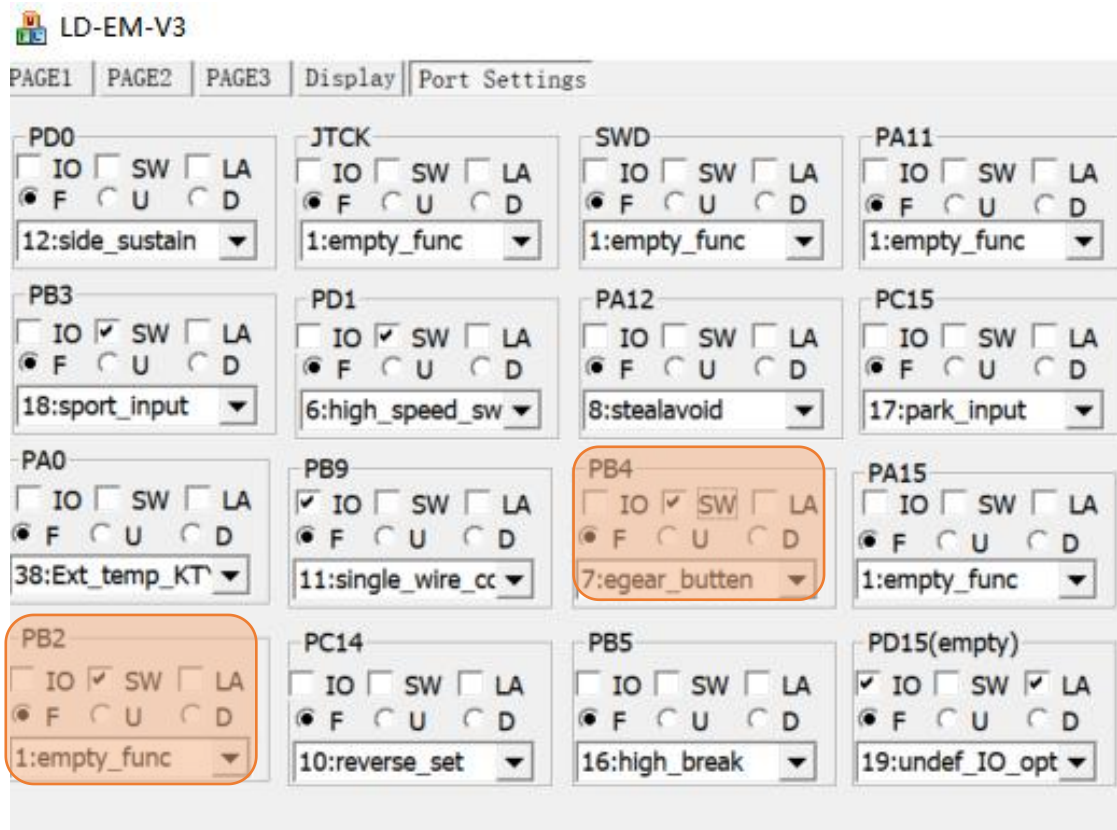


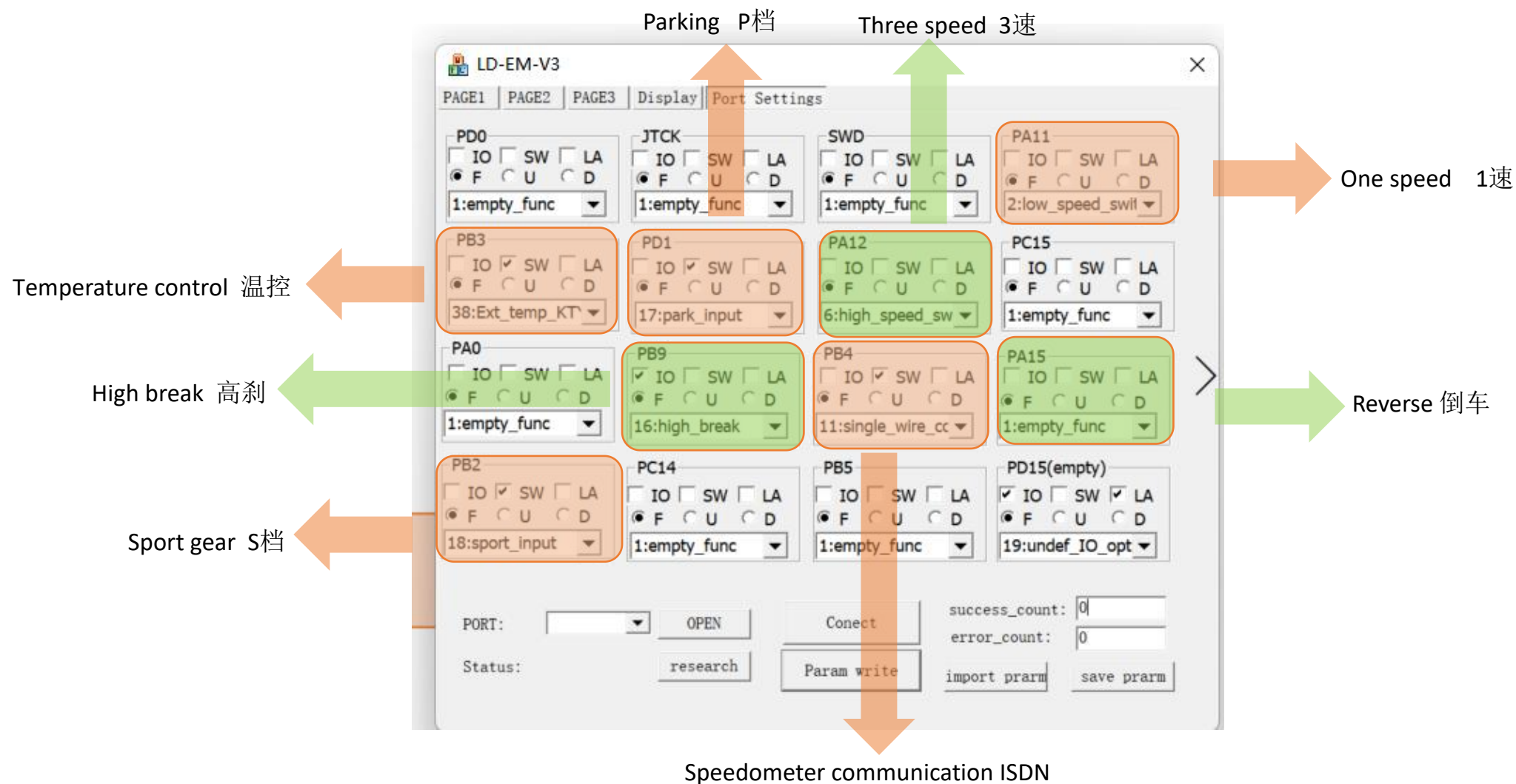


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BUTTON THRESS SPEED SETTING 点动三速设置

(EM150SP after 20200321) in common use





Thank you
for using